In vivo studies of Metadoxine, NAC, Neem and Burdock Root on CCl4-induced Toxicity on Swiss Albino mice *(Mus musculus)*

Thesis submitted for the fulfillment of the degree of

Bachelor of Technology in Biotechnology

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TO WHOM IT MAY CONCERN

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ABSTRACT

Metadoxine, N-acetylcysteine (NAC), Neem and Burdock Root are commonly used in clinical settings for their hepatoprotective and antioxidant properties. Metadoxine, a derivative of the amino acid methionine, is known to aid in liver detoxification, while NAC is a precursor to glutathione, which plays a key role in reducing oxidative stress and promoting cellular repair. On the other hand Neem (Azadirachta indica) stabilizes hepatocyte membranes and preserves cellular ultrastructure, especially mitochondria and endoplasmic reticulum. Its active compound, nimbolide, not only reduces elevated liver enzymes (AST, ALT, ALP) but also restores total protein and albumin levels. Burdock root (Arctium lappa) restores cytochrome P-450 enzyme activity and reduces oxidative stress through potent antioxidant mechanisms. Its polysaccharides and active compounds suppress lipid peroxidation, lower malondialdehyde levels, and inhibit inflammatory pathways, particularly the MAPK pathway. This results in decreased hepatocyte apoptosis and improved liver function, Despite their therapeutic potential, the effects of these compounds on liver function and their potential toxicity require further investigation. Swiss Albino mice (Mus musculus) are an ideal animal model for such studies, owing to their gene homology to humans. The liver, as the primary organ for drug metabolism, is the focus of this study, which aims to examine the hepatotoxic effects of Metadoxine, NAC, Neem and Burdock Root in a dosage-response relationship manner. We wish to assess the activities of hepatic diagnostic enzymes, including Alkaline Phosphatase (ALP), Acid Phosphatase (ACP), SGPT and SGOT, to evaluate the liver function. Additionally, chromosomal aberrations and other cytotoxic effects are also being studied with varying dosages and exposure durations. The findings from this research can provide critical insights into the safety and therapeutic efficacy of these drugs, contributing to a better understanding of their potential toxic effects on the human body.

Keywords: *Metadoxine, N-acetylcysteine, Neem, Burdock Root hepatotoxicity, antioxidant, Acid Phosphatase, Alkaline Phosphatase, SGPT, SGOT, cytotoxicity.*

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1.) INTRODUCTION

Metadoxine supports liver health by accelerating recovery in alcoholic fatty liver disease, improving liver function tests, and reducing steatosis, even in patients who continue drinking. It enhances short-term survival in severe alcoholic hepatitis when added to standard treatment, lowers complications like encephalopathy, and shows antioxidant and hepatoprotective effects by reducing oxidative stress and fat accumulation in the liver. Its benefits are most evident in alcohol-related liver conditions. [Tijera *et al.,*2014].

N-acetylcysteine (NAC) plays a vital role in replenishing glutathione, a key antioxidant that protects liver cells from oxidative stress and injury. It is well-established for treating acetaminophen-induced liver failure and also shows benefits in non-acetaminophen liver injuries, including non-alcoholic fatty liver disease, by improving liver function tests and reducing inflammation. NAC's antioxidant and anti-inflammatory effects help prevent or reduce liver damage from various toxins and conditions. [Vemuri *et al*.,2020].

Neem supports liver health primarily through its antioxidant and anti-inflammatory properties, which help reduce oxidative stress and protect liver cells from damage. Animal studies show neem leaf extract can decrease liver enzymes like AST and ALT, indicating reduced liver injury, and may protect against drug-induced liver damage. Neem also promotes bile secretion and aids detoxification, further supporting liver function. However, most evidence comes from animal studies, and human data are limited. [Seriana *et al.,*2021].

Burdock root uniquely supports liver health by providing chlorogenic acid, which enhances the liver's ability to metabolize fats, reduces fat buildup, and protects liver cells from damage. It stimulates bile production, aiding detoxification, and helps regulate blood sugar, reducing liver strain. Studies show burdock root can lower liver enzymes linked to damage, offering protective effects against conditions like fatty liver disease and potentially even inhibiting liver cancer cell growth. [Sheng W *et al*.,2023]

1.1 Historical Background of Metadoxine, NAC, Neem, Burdock Root:

Metadoxine, chemically known as pyridoxine-pyrrolidone carboxylate, was first developed in the late 20th century for managing acute and chronic alcohol intoxication and alcoholic liver disease. It quickly gained approval in several countries, including China in 2005, due to its ability to accelerate alcohol clearance from the blood and improve clinical outcomes in intoxicated patients. Over time, research expanded to demonstrate its efficacy in improving abstinence rates and survival in severe alcoholic hepatitis. In the 2010s, metadoxine also

attracted interest for neuropsychiatric uses, particularly attention deficit hyperactivity disorder (ADHD), leading to the development of extended-release formulations and clinical trials in the U.S. and Europe. [Tijera *et al*.,2015]

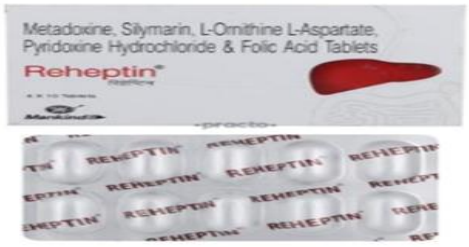


Figure 1: Metadoxine tablets



Fig 2: N-actylcysteine (NAC)

N-acetylcysteine (NAC) is an acetylated form of the amino acid cysteine and serves primarily as a precursor for glutathione (GSH), a critical intracellular antioxidant. Histologically, NAC's background is rooted in its ability to reduce oxidative stress and inflammation in tissues. It acts by donating cysteine for GSH synthesis, directly scavenging reactive oxygen species (ROS), and reducing disulfide bonds, which is the basis for its mucolytic action in pulmonary diseases. In tissue studies, NAC has shown protective effects against cellular damage by restoring normal histological architecture, such as promoting remyelination in the brain and protecting glandular tissues from metabolic stress. Its anti-inflammatory and cytoprotective effects are evident in various organs, where it inhibits neutrophil activation, reduces apoptosis, and modulates immune responses. [Sahasrabudhe *et al*., 2023]



Figure 3: Neem Plant

Neem (*Azadirachta indica*) has a rich histological profile linked to its medicinal use. Studies in animal models show that prolonged administration of neem leaf extract can induce histopathological changes in organs such as the spleen, liver, and kidneys. Notably, these changes include central vacuolation and lymphocyte accumulation in the spleen, hypertrophy of the central vein in the liver, and shrinking of the glomeruli with lymphocyte accumulation in the kidney. Despite these alterations, short-term use generally does not cause significant histological disruption. Its bioactive compounds exhibit anti-inflammatory, antioxidant, and wound-healing properties, supporting tissue repair and reducing inflammation through modulation of immune cell activity. [Alzohairy *et al*., 2016]



Figure 4: Burdock Root

Burdock root (*Arctium lappa*) has a long history as both food and medicine, originating in Europe and Asia and later spreading to North America. Its use dates back thousands of years, with records in traditional Chinese, Ayurvedic, and Native American medicine, where it was valued for detoxification, blood purification, and treating various ailments. In medieval Europe and Asia, burdock root was prescribed for conditions like constipation, arthritis, and skin diseases, and Native Americans used the entire plant for nutrition and healing. The plant's distinctive burrs even inspired the invention of Velcro in the 20th century. [Sheng W *et al*.,2023]

1.2 Mechanism of Action of Metadoxine

Metadoxine is Pyridoxine pidolate. Metadoxine has a direct effect on alcohol metabolism. It accelerates the transformation of alcohol into acetaldehyde and is excreted in urine. It aids in preventing liver damage resulting from prolonged alcohol intake and helps in reversing fatty liver. Antifibrotic agent: In hepatic stellate cells, Metadoxine prevents the collagen synthesis & reduces fibrosis and also is a synthetic antioxidant, provides stronger antioxidant protection. Metadoxine, prevents the redox imbalance and tumor necrosis factor (TNF-a) one of the earliest events in hepatic damage. It increases the cerebral release of GABA and acetylcholine and the cerebral levels of ATP. Therefore is an antagonist of alcohol metabolic effects and helps to restore normal neuropsychological behaviour. [Tijera *et al.,*2014]

1.2.1 Physical & Chemical Properties of Metadoxine:

● Molecular Weight: 298.29 g/mol

● ATC Code: A05BA09 (WHO)

● IUPAC Name: 4,5-bis(hydroxymethyl)-2-methylpyridin-3-ol;(2S)-5 oxopyrrolidine-2-carboxylic acid

● Chemical Formula: C13H18N2O6

● Molecular Formula: C8H11NO3.C5H7NO3

● CAS Number: 74536-44-0

● Melting Point: 102-104°C

● Solubility: DMSO (Slightly), Methanol (Slightly)

● Appearance: Solid

● Colour: White to Light Yellow

● Storage Condition: Sealed in dry,2-8°C

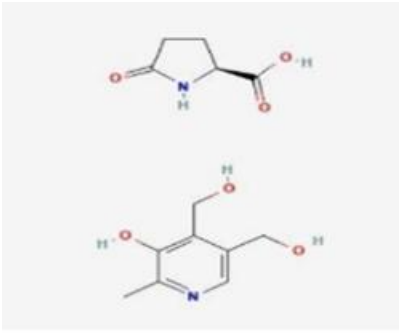


Figure 5: Chemical structure of Metadoxine

1.2.2 Pharmacokinetics of Metadoxine

● Absorption: It is readily absorbed after oral administration.

● Distribution: It is mainly distributed in the body in protein bound form. It is stored primarily in the liver and to a lesser extent muscle and brain.

● Metabolism: It is metabolised in the liver.

1.2.3 Side Effects of Metadoxine:

Neuropathy ; Unstable gait ; Drowsiness; Perioral numbness ; Paresthesia; Numbness of feet Somnolence; Sensory neuropathy; Ataxia

The most common side effects associated with Metadoxine include nausea, upset stomach and diarrhoea.

1.3 Mechanism of action of N-Acetylcysteine:

NAC exerts its therapeutic effect in APAP overdose through several mechanisms. APAP metabolism in therapeutic dosing primarily occurs through glucuronidation and sulfation (>90%), with less than 5% being oxidized by CYP450 isoform (predominately CYP2E1)

to produce a toxic metabolite called N-acetyl-p-benzoquinone imine (NAPQI), which is the precursor to cellular injury. Glutathione in the liver can normally detoxify these minuscule quantities of NAPQI and prevent tissue damage. In APAP overdose, glucuronidation and sulfation pathways are saturated, and the CYP450 pathway has more significance, producing toxic metabolites that deplete the glutathione reserves, leading to their accumulation and subsequent tissue injury by binding to cellular macromolecules. NAC repletes glutathione reserves by providing cysteine, an essential precursor in glutathione production. NAC itself also binds to toxic metabolites and scavenges free radicals. NAC also increases oxygen delivery to tissues, increases mitochondrial ATP production, and alters the microvascular tone to increase blood flow and oxygen delivery to the liver and other vital organs. In COPD, cystic fibrosis, and other lung conditions, nebulized NAC has mucolytic, anti-inflammatory, and antioxidant properties. Studies are ongoing to understand NAC's therapeutic efficacy, ideal dose ranges, and the most effective mode of drug delivery for these indications. [Sahasrabudhe *et al*., 2023].

1.3.1. Physical & Chemical Properties of NAC:

• Molecular Formula: C5H9NO3S

• Molar Mass: 163.19 g·mol−1

• IUPAC: N-acetyl-L-cysteine

• Melting Points: 109–110-degree celsius

• Form: WHITE, CRYSTALLINE POWDER

• Odor: Slight Acetic Odor

• Solubility: >24.5 [ug/mL] ; Taste: Sour Taste

• Stability / Shelf Life: Stable in ordinary light; stable at temp up to 120 °C; nonhygroscopic (oxidizes in moist air)

• Optical Rotation Specific optical rotation: +5 deg at 20 °C (concn = 3 g/100 mL water)

• pH: 2 TO 2.75 (1 IN 100 ML)

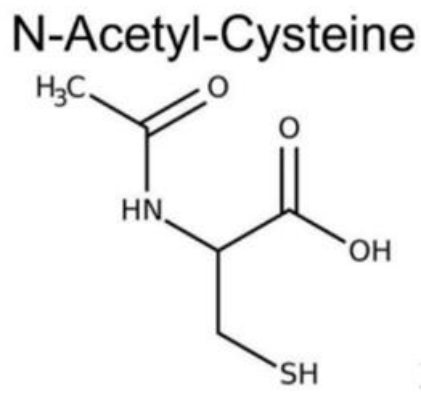


Figure 6: Molecular Structure of NAC

1.3.2. Pharmacokinetics of NAC:

• Absorption: The time taken to reach maximum blood concentration (Tmax) for inhalation through a nebulizer is 1 to 2 hours; for oral administration, an effervescent tablet is around 2 hours.

• Distribution: When administered intravenously (IV), NAC tends to attain a steady state distribution (Vdss) of about 0.47%. The plasma protein binding of NAC is between 66% to 87%.

• Metabolism: Biotransformation of NAC leads to the formation of metabolites, disulfide, cysteine, and conjugates (N, N-diacetyl cysteine, N-acetylcysteine cysteine, N-acetylcysteine-protein, etc). Further metabolism of cysteine metabolites leads to the formation of glutathione and other metabolites.

• Elimination: The termination half-life of NAC after a single IV dose is 5.6 hours. At the same time, NAC is cleared renally at a rate of 0.11 L/hr/kg, constituting about 30% of total body clearance.

1.3.3. Side-effects of NAC:

Oral NAC may cause nausea, vomiting, diarrhea, flatus, and gastroesophageal reflux. IV NAC can cause rate-related anaphylactoid reactions in up to 18% of patients, Most anaphylactoid reactions are mild (6%) or moderate (10%), with severe reactions like bronchospasm and hypotension rare at 1%. Interestingly, anaphylactoid reactions occur more commonly with

lower APAP levels. One possible explanation is that APAP decreases the histamine release from mast cells and mononuclear cells proportionate to the dose ingested. Bronchospasm more commonly occurs in patients with pre-existing reactive airway diseases, like asthma. Bronchodilating agents are effective in treating these patients.

1.3.4. Toxicity of NAC:

Given the drug's complicated regime, NAC has a high potential for iatrogenic errors, including overdose.The signs of overdose are documented to include hemolysis, thrombocytopenia, metabolic acidosis, acute renal failure, and elevated serum bilirubin. Massive accidental NAC administration of 100 mg/kg/hr had resulted in cerebral edema, seizures, uncal herniation, and permanent brain injury in a patient with an APAP overdose.

1.4. Mechanism action of Neem

Neem (Azadirachta indica) acts through multiple mechanisms due to its rich content of active compounds like azadirachtin, nimbolide, nimbin, and quercetin. Its antimicrobial action involves disrupting microbial cell walls and inhibiting growth. Neem's antioxidant properties enable it to scavenge free radicals, reducing oxidative stress and preventing cellular damage. In metabolic diseases, neem upregulates antioxidant enzymes via Nrf2, enhances nitric oxide for vasodilation, and blocks calcium channels, contributing to blood pressure regulation. For diabetes, neem increases GLUT4 expression, improving glucose uptake and insulin signaling. In cancer, neem modulates tumor suppressor genes (p53, pTEN), inhibits angiogenesis (VEGF), and induces apoptosis by activating pro-apoptotic proteins and suppressing anti-apoptotic genes. Neem also shows antiviral and insecticidal activity by interfering with viral replication and disrupting insect development through azadirachtin. [Alzohairy *et al*., 2016]

1.4.1. Physical and Chemical Properties of Neem:

- Molecular Formula: $C_{35}H_{44}O_{16}$ (azadirachtin A).
- Melting Points: Decomposes >200°C (azadirachtin A); 92°C (extract).
- Form: Liquid (seed/fruit oil) or solid (flesh oil).
- Odor: Peanut-garlic combination.
- Solubility: 10.7 g/L in water (azadirachtin A at pH 7); Taste: bitter taste.
- Stability: pH stable for 30 days in nanoemulsion; photolysis half-life ranges 5.5 days (summer) to 1.8 months (winter)**.**

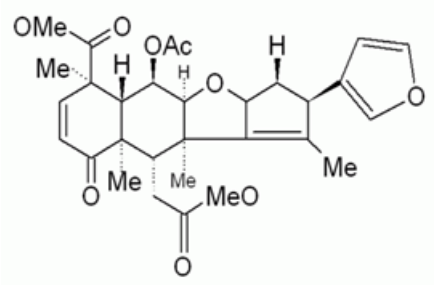


Figure 7: Chemical Structure of Neem

1.4.2. Pharmacokinetics of Neem:

- Absorption: Neem compounds generally show variable gastrointestinal (GI) absorption. Some, like nimbolide, have low oral bioavailability, while others may have higher absorption depending on their chemical structure and solubility.
- Distribution: Distribution varies by compound. Some neem phytochemicals can cross the blood-brain barrier, but most do not. Lipophilicity and water solubility influence their ability to circulate and reach target tissues**.**
- Metabolism: Neem components are metabolized primarily by hepatic enzymes, including cytochrome P450s. Some compounds may inhibit CYP enzymes (e.g., CYP1A2, CYP3A4), potentially affecting the metabolism of other drugs.
- Elimination: Elimination pathways include renal and biliary excretion. Neem compounds are generally eliminated after hepatic metabolism, but specific rates and routes depend on the individual molecule's properties.

1.4.3. Side Effects of Neem

Neem can cause serious side effects, especially in children, including vomiting, diarrhea, drowsiness, seizures, loss of consciousness, coma, and even death after ingesting neem oil or seeds. In adults, high doses or prolonged use may harm the liver and kidneys. Neem may worsen autoimmune diseases, lower blood sugar excessively, reduce fertility, and interfere with anti-rejection medications in transplant patients. Allergic reactions, such as dermatitis, have also been reported. Neem is unsafe during pregnancy and breastfeeding due to miscarriage risk.

1.5. Mechanism Action of Burdock Root:

Burdock root (Arctium lappa) acts through multiple mechanisms due to its diverse bioactive compounds. Its lignans, such as arctiin and arctigenin, exert anti-inflammatory effects by inhibiting pro-inflammatory mediators like nitric oxide (NO), tumor necrosis

factor-alpha (TNF-α), and interleukin-6 (IL-6), as well as suppressing the NF-κB pathway. Burdock root is rich in antioxidants, including quercetin, luteolin, and phenolic acids, which scavenge free radicals and protect cells from oxidative stress, contributing to its anti-inflammatory and anticancer effects. It contains inulin, a prebiotic fiber that supports gut health and helps regulate blood glucose by acting on cell surface receptors and inhibiting alpha-glucosidase activity, which slows carbohydrate absorption and may benefit diabetes management. Burdock root also promotes detoxification by stimulating lymphatic drainage and diuresis, aiding in the removal of toxins from the body. Some compounds, like tannins, induce macrophage responses and modulate immune function, while others, such as polyacetylenes and chlorogenic acid, provide antimicrobial benefits. [Mondal SC *et al*., 2022].

1.5.1. Physical and Chemical Properties of Burdock Root:

- Molecular Formula:(C6H10O5).
- Melting Points: typically above 150°C.
- Form: Burdock root is solid, fibrous, and fleshy when fresh; it can also be found dried and powdered.
- Odor: It has an earthy, slightly sweet, and herbal odor, characteristic of root vegetables.
- Solubility: Contains water-soluble polysaccharides (inulin, pectin) and phenolic compounds.
- Taste: Mildly sweet, earthy, and slightly bitter.
- Stability: Stable when kept dry and protected from light; bioactive compounds may degrade with prolonged exposure to heat, moisture, or air.

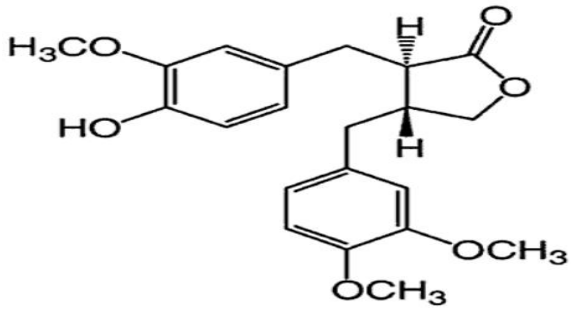


Figure 8: Chemical Formula of Burdock Root

1.5.2. Pharmacokinetics of Burdock Root:

- Absorption: Key active compounds in burdock root, such as arctigenin and arctiin, are absorbed after oral administration. Studies in animals show that arctigenin from burdock root powder is rapidly absorbed, reaching peak plasma concentration in under an hour.
- Distribution: After absorption, arctigenin is widely distributed in the body, but its tissue distribution is generally low following intravenous administration. Oral administration leads to broader distribution compared to intravenous, possibly due to slower elimination.
- Metabolism: Arctigenin and arctiin undergo extensive metabolism, especially glucuronidation and hydrolysis in the liver, intestine, and plasma, which can limit their bioavailability and efficacy after oral intake.
- Elimination: Elimination of arctigenin is relatively rapid after intravenous administration (elimination half-life ~3 hours in animal studies), but much slower after oral administration (elimination half-life up to 63 hours), suggesting sustained effects with oral dosing.

1.5.3. Side Effects of Burdock Root:

Burdock root is generally safe when eaten as food, but possible side effects include allergic reactions (especially in people allergic to ragweed, daisies, or chrysanthemums), rash or contact dermatitis, and rare cases of anaphylaxis. It may slow blood clotting, increasing bleeding risk, and acts as a diuretic, potentially causing dehydration. People with diabetes should use caution, as it may lower blood sugar. Pregnant or breastfeeding women should avoid it due to insufficient safety data.

2. LITERATURE REVIEWS:

2.1) Toxicological Profile of Carbon Tetrachloride: Carbon tetrachloride (CCl4) is a colorless, volatile, non-inflammable liquid that is produced by the mixture of chlorine with chloroform in the presence of light. It is structurally a chlorinated hydrocarbon called tetrachloromethane (International Union of Pure and Applied Chemistry nomenclature/IUPAC name). CCl4 toxicity develops not due to the CCl4 itself but for the generation of free radical CCl3 and other metabolites produced by cytochrome P450. Ultimately they lead to cellular damage by alteration of cellular structure through lipid peroxidation and in some other pathways. Severe conditions may develop through multiple organ dysfunction by these free radicles [Rockmeal RO,1983]

2.2) Involvement of CCl4 in Liver Toxicity: Carbon tetrachloride(CCl4) induced cellular damage may result from either covalent bond formation between reactive intermediates and cellular components or from enhanced lipid peroxidation triggered by free radical intermediates. It causes intracellular and intramembranous lipid destruction.[Weber LW, 2001]. In the liver, the metabolism of CCl4 to other toxic metabolites is catalyzed by cytochrome P450, especially by its isoenzyme CYP2E1. This enzyme is also a part of the hepatic microsomal ethanol-oxidizing system (MEOS), which is involved in the hepatic metabolism of ethanol. Both of these enzymes are inducible by prolonged alcohol use, and even a single dose of alcohol may induce MEOS activity. That's why an alcoholic person develops severe manifestation if associated with CCl4 toxicity. The break-down products such as reactive aldehydes lead to further damage like increased membrane permeability, which is one indicator of impending cell death. Free radical formation of CCl4 induced hepatotoxicity depends on the partial pressure of oxygen. CCl3* and CCl2* radicals form in low partial pressure that leads to covalent metabolite binding, which mostly affects the metabolism of lipids(decreased transport out of the hepatocyte, increased synthesis) and ultimately causes steatosis or fatty liver. In contrast, CCl3- OO* radical forms in high oxygen partial pressure with consequent lipid peroxidation that push the cell from steatosis to apoptosis. Hepatocyte protein synthesis is also suppressed by CCl4, which leads to loss of structural and functional integrity of cells. Another product of CCl4 metabolism is phosgene that might be another cause of hepatotoxicity.[Teschke R. 2018].

2.3) Effect of CCl4 in liver damage of Albino mice, an experimental study: In this present study we have taken 6 albino mice, each of them where injected with a dosage of 0.25 ml of 10% CCl4 in a 2 ml syringe to estimate biochemical and chromosomal toxicity markers by studying phosphatase enzyme level, LFT enzyme levels and chromosomal aberrations

2.4) Metadoxine as a hepatoprotective drug: Metadoxine (pyridoxine pyrrolidone carboxylate) is an ion pair between pyridoxine (vitamin B6) and pyrrolidone carboxylate. Metadoxine is considered to be a beneficial treatment for alcoholic liver disease. It accelerates the clearance of ethanol and acetaldehyde from the body and diminishes the damaging effect induced by ethanol.[7] Metadoxine is effectively able to preserve GSH reservoir and prevents lipid peroxidation, collagen disposition, and tumor necrosis factor-alpha (TNF-α) secretions induced by ethanol and acetaldehyde in hepatocytes and hepatic stellate cells in culture. Experimental studies in ethanol-administered rats have indicated that metadoxine restores liver GSH content and prevents reduction of hepatic adenosine triphosphate (ATP) concentration caused by ethanol. Moreover, metadoxine may be useful for treating nonalcoholic steatohepatitis due to inhibition of causative factors that are involved in oxidative stress.In CCl4-chronic administration model of liver fibrosis, animals treated with CCl4 and metadoxine concurrently had less severe fibrosis and inflammation with normal serum levels of immunoreactive prolyl hydroxylase in comparison to rats treated only with CCl4. Therefore, this study for the first time was aimed to examine the probable protective effects of metadoxine in hepatotoxicity induced by acetaminophen in mice, and the mechanisms are involved.[Shpilenya *et al*., 2002]

2.5) Hepatoprotective Effect of Metadoxine on CCl4-induced Liver Toxicity in Albino Mice: Metadoxine, a pyridoxine derivative, has demonstrated hepatoprotective properties in various experimental models, including CCl4-induced liver injury. Antioxidant activity where metadoxine can scavenge free radicals, such as reactive oxygen species (ROS) and reactive nitrogen species (RNS), thereby reduces oxidative stress. Another is Anti-inflammatory effects here metadoxine can inhibit the production of inflammatory cytokines, such as tumor necrosis factor-alpha (TNF-α) and interleukin-1β (IL-1β), thus reducing inflammation. By suppressing the inflammatory response, metadoxine can limit the extent of liver injury and promote tissue repair. Finally, hepatoprotective effects include metadoxine,

which can protect hepatocytes from damage, promote cell regeneration, and inhibit fibrosis. It can also modulate various signaling pathways involved in liver injury, such as the nuclear factor-kappa B (NF-κB) pathway and the mitogen-activated protein kinase (MAPK) pathway. To investigate the hepatoprotective effects of metadoxine against CCl4-induced liver toxicity, researchers often employ animal models, such as albino mice. CCl4 is administered to mice, typically through intraperitoneal (IP) injection, to induce liver injury. Metadoxine is administered orally at various doses and frequencies. The efficacy of metadoxine is assessed by evaluating biochemical parameters, such as serum levels of liver enzymes (ALT and AST), histopathological examination of liver tissue, and measurement of oxidative stress markers and inflammatory cytokines..[Mazraati P *et al.,*2018]

2.6) Experimental study of Metadoxine for the treatement of liver damage in albino mice: In this study, we have taken four groups of albino mice where one group was injected with only CCl4 and the other group was given CCl4 first and kept for seven days observation and then metadoxine where given in three different dosage (dosage include 250, 500, 750 gm) to estimate biochemical and chromosomal toxicity markers by studying phosphatase enzyme level, LFT enzyme levels and chromosomal aberrations

2.7) NAC an-anti liver toxic drug: The liver significantly contributes to bodily functions through the production of various proteins and enzymes. Liver enzymes, specifically aspartate aminotransferase (AST) and alanine aminotransferase (ALT), serve as surrogate markers for liver damage; an increase in their serum levels results from liver damage and hepatocyte injury. Consequently, this elevation indicates liver cell destruction. AST and ALT are the most critical liver enzymes associated with increased liver damage. Additionally, elevated levels of alkaline phosphatase (ALP), an enzyme secreted by the liver, also signify potential liver issues. The levels of bilirubin, albumin, and platelets in the blood, which indicate liver cell functionality, may be altered in cases of liver dysfunction. Oxidative stress, lipid peroxidation, inflammation, and impaired immune response represent the primary processes involved in the pathogenesis of liver disorders. Therefore, chronic diseases with similar pathogenesis may elevate liver enzymes and impair liver function. Previous studies have shown that, alongside healthy diets and lifestyle modifications, antioxidant and anti-inflammatory supplements may assist in the treatment of liver dysfunction. One such supplement with both antioxidant and anti-inflammatory properties for treating acute liver, kidney, and lung disorders is N-acetylcysteine (NAC). NAC is a derivative of L-cysteine and serves as a precursor to

glutathione (GSH). GSH is acknowledged as a potent antioxidant that safeguards cells from oxidative damage. As a result, NAC may function as a potential agent in enhancing antioxidant capacity. No side effects have been reported for NAC supplements at doses up to 3 g/day, as it is rapidly absorbed in both animal and human subjects following oral administration. Recent studies suggest a potentially beneficial effect of NAC on liver function in humans. For instance, NAC significantly reduced the Non-Alcoholic Fatty Liver Disease Activity Score (NAS) and serum/plasma ALT concentration in patients with Non-Alcoholic Steatohepatitis (NASH). While the beneficial effects of NAC on liver, kidney, and lung function across various conditions have been emphasized, the varying doses of NAC administered in the treatment of a wide array of disorders necessitate further investigation into its efficacy. Given the controversial findings regarding the effect of NAC on liver function and the absence of a comprehensive systematic review or meta-analysis, the present study seeks to summarize the impact of NAC on liver function and ascertain its magnitude and direction. (Galicia M et al., 2024)

2.8) Treatment of liver damage by CCl4 of albino mice by using NAC: NAC replenishes intracellular glutathione levels, which are crucial for cellular detoxification and antioxidant defense. Furthermore, NAC exhibits anti-inflammatory properties. It can inhibit the production of pro-inflammatory cytokines, such as tumor necrosis factor-alpha (TNF-α) and interleukin-1β (IL-1β), which play a significant role in the inflammatory response associated with CCl4-induced liver injury. By reducing inflammation, NAC can limit the extent of tissue damage and promote liver regeneration. Several studies have demonstrated the efficacy of NAC in protecting against CCl4-induced liver injury in animal models. Treatment with NAC has been shown to significantly reduce serum levels of liver enzymes, such as alanine aminotransferase (ALT) and aspartate aminotransferase (AST), which are markers of liver damage. Histopathological examination of liver tissue from NAC-treated animals has revealed reduced inflammation, necrosis, and fibrosis compared to untreated animals. [Tenório MC *et. al.*, 2021]

2.9) Use of NAC against CCl4 toxicity in liver of albino mice, an experimental study: In this study, we have taken four groups of albino mice where one group was injected with only CCl4 and the other group was given CCl4 first and kept for seven days observation and then NAC where given in three different dosage (dosage 300, 600, 750 gm) to estimate

biochemical and chromosomal toxicity markers by studying phosphatase enzyme level, LFT enzyme levels and chromosomal aberrations.

2.10) Synergistic study between two anti-liver toxic drug: In this study we have taken two groups of albino mice where one group was injected with only CCl4 and the other group was given CCl4 first and kept for seven days observation and then NAC and metadoxine where given at the same time (dosage include NAC 600mg + metadoxine 500 mg) to estimate biochemical and chromosomal toxicity markers by studying phosphatase enzyme level, LFT enzyme levels and chromosomal aberrations.

2.11) Neem as an anti-liver toxic drug: Neem (Azadirachta indica) acts as an effective anti-liver toxic agent primarily due to its potent antioxidant and anti-inflammatory properties. Experimental studies in animal models have demonstrated that neem leaf extracts significantly reduce liver damage caused by various toxins such as rifampin, paracetamol, and carbon tetrachloride. Neem administration lowers elevated liver enzymes like AST, ALT, and ALP, which are markers of hepatocellular injury, and restores them toward normal levels. It also decreases pro-inflammatory cytokines (e.g., TNF-α, IL-1β), thereby reducing inflammation and hepatocyte necrosis. Histopathological examinations reveal that neem preserves normal liver architecture by preventing sinusoidal congestion, fatty changes, and cell apoptosis. The hepatoprotective effect is largely attributed to neem's phytochemicals, including flavonoids and other antioxidants, which scavenge free radicals and reduce oxidative stress,the main mechanism underlying toxin-induced liver injury. While some reports note potential toxicity at very high doses, typical therapeutic doses of neem extracts have shown safety and efficacy in protecting the liver from chemical-induced damage in preclinical studies. However, further clinical trials in humans are necessary to confirm these benefits and establish standardized dosing. [Okojie S *et al*., 2017]

2.12) Hepatoprotective Effect of Neem Against CCl4-Induced Liver Toxicity:

The primary function of neem, particularly its active constituents such as nimbolide and azadirachtin-A, is to mitigate oxidative stress generated by the metabolism of CCl4, which produces deleterious free radicals that lead to lipid peroxidation and damage to hepatocytes. Neem enhances the liver's antioxidant defense by elevating levels of glutathione (GSH) and superoxide dismutase (SOD), thereby diminishing lipid peroxidation and stabilizing cellular membranes. This antioxidative effect averts structural damage to hepatocytes, maintains the integrity of organelles such as mitochondria and the rough endoplasmic reticulum, and reduces

inflammation by downregulating pro-inflammatory cytokines. Neem also normalizes elevated liver enzyme levels and improves markers of protein synthesis, such as albumin, indicating a restoration of liver function. In summary, the hepatoprotective mechanism of neem encompasses the scavenging of free radicals, stabilization of cellular membranes, and promotion of liver tissue regeneration, collectively protecting and restoring liver function against CCl4 toxicity in a dose-dependent manner comparable to standard pharmacological agents such as silymarin. [Sunday GA *et al*.,2024]

2.13) Use of Neem against CCl4 toxicity in liver of albino mice, an experimental study: In this study, we have taken four groups of albino mice where three group was first injected with CCl4 then kept for seven days after which different doses of neem were given to three different groups of mice (doses include 0.1,0.2,0.4 gm), another synergistic study was done including CCl4 which was given first and kept for seven days after which NAC and neem were given. Both NAC and neem were given at the same time (dosage for NAC 600 mg and neem 0.2 gm) to estimate biochemical and chromosomal toxicity markers by studying phosphatase enzyme level, LFT enzyme levels and chromosomal aberrations.

2.14) Burdock Root an antitoxic liver drug: Burdock root is rich in bioactive compounds such as polyphenols, flavonoids (e.g., quercetin), chlorogenic acid, and inulin (a prebiotic fiber) that not only scavenge free radicals and reduce oxidative stress but also support gut health and immune function. Its ability to stimulate bile production enhances the liver's natural detoxification pathways, promoting efficient elimination of toxins and metabolic waste. It also exhibits hypolipidemic activity, helping to regulate blood lipid levels and reduce fatty liver risk, which is not a common feature in many standard hepatoprotective drugs. Burdock root supports lymphatic drainage and acts as a natural diuretic, facilitating toxin clearance beyond the liver, which enhances its detoxifying impact. This combination of liver-specific and systemic detoxification, along with gut microbiome support and metabolic regulation, makes burdock root uniquely comprehensive in its hepatoprotective action compared to other drugs that often target only one or two of these pathways. [Shyam M *et al.,*2024]

2.15) Hepatoprotective effect of burdock root over CCl4 inducing liver toxicity: Burdock root significantly restores the liver's antioxidant defense by increasing glutathione content and NADPH-Cytochrome-c reductase activity while reducing lipid

peroxidation and serum transaminase levels caused by CCl4 toxicity. It also helps to maintain cytochrome P-450 content, which is crucial for detoxification processes impaired by CCl4. This broad enhancement of the liver's intrinsic antioxidant enzymes and detoxifying systems reduces oxidative stress and cellular damage more comprehensively. Modulation of immune responses and inhibition of inflammatory mediators, adds an immunomodulatory dimension to its hepatoprotection. [Mahboubi M et al., 2021].

2.16) Use of Burdock against CCl4 toxicity in liver of albino mice, an experimental study: In this study, we have taken four groups of albino mice where three group was first injected with CCl4 then kept for seven days after which different doses of burdock root were given to three different groups of mice (doses include 0.1,0.2,0.4/ml), another synergistic study was done including CCl4 which was given first and kept for seven days after which metadoxine and burdock root were given. Both metadoxine and burdock root were given synergistically (dosage for metadoxine 500 mg and burdock root 0.1/gm) to estimate biochemical and chromosomal toxicity markers by studying phosphatase enzyme level, LFT enzyme levels and chromosomal aberrations.

3. AIM AND OBJECTIVE:

1) To study the toxic effect of CCl4 causing liver damage in Swiss Albino Mice (Mus musculus).

2) In vivo studies on hepatoprotective effect of Metadoxine on CCl4 induced liver toxicity in Swiss Albino mice (Mus musculus).

3) In vivo studies on hepatoprotective effect of NAC on CCl4 induced liver toxicity in Swiss Albino mice (Mus musculus).

4) To study the trend of dose response of Metadoxine and NAC on mice model and corroborate it for human applications.

5) In vivo studies on hepatoprotective effect of Neem on CCl4 induced liver toxicity in Swiss Albino mice (Mus musculus).

6) In vivo studies on hepatoprotective effect of Burdock Root on CCl4 induced liver toxicity in Swiss Albino mice (Mus musculus).

7) To study the trend of dose response of Neem and Burdock Root on mice model and corroborate it for human applications.

8) Synergistic study of two synthetic drug and two natural drug on CCl4 induced liver toxicity in Swiss Albino mice and corroborate it for human application.

4. Materials and methods:

4.1 Animal Model:

- Healthy male Swiss Albino mice (*Mus musculus*) were procured from a registered breeder, Ghosh Scientific, Kolkata, India and reared in animal cages.
- All animals were acclimatized at laboratory conditions (temperature of 25 ± 1°C. repetitive 12h light/dark cycles) for 1 week before the commencement of the study. They were fed with a standard meal of powdered ground pulses (Sattu) and drinking water ad libitum.
- The mice were randomly divided into different experimental groups.



Fig 8: Swiss Albino Mouse (Male)

4.2 Standard diet:

- Standard diet for mice was prepared by mixing gram flour (sattu) with water until a dough was formed. The dough was fed to the mice.
- Normal drinking water was also provided to the mice in feeding bottles.

4.3 Chemicals required:

- Para-nitro phenol phosphate, (PNPP) (Merck, USA)
- 0.1 M Acetic Acid acetate buffer (pH 5)
- 0.1 M NaOH glycine buffer (pH 9)
- 1 N NaOH
- 0.1 N KCl
- Para-nitro phenol (PNP) (Merck)
- Distilled/RO water
- Chloroform (Emplura)
- Ethanol (Chongshu Hongshei Co. Ltd.)
- Methanol (Emplura)
- Glacial Acetic Acid (Emplura)
- Giemsa stain (Himedia)
- SGPT kit
- SGOT kit

4.4 Instruments Required:

- Tissue homogenizer (Borosil, India)

- Centrifuge (Remi)
- Visible Spectrophotometer (Labman)
- Dissection kit (Falcon, India)
- Centrifuge tubes (15 ml respectively)
- Optical Microscope (Magnus opto)
- Glass slides
- Pipettes: Glass pipette (5 mL)
- Spirit lamp
- 1 mL syringe
- Test tubes

4.5 Drug Administration:

- The mice were kept in optimal laboratory conditions (temperature of 25 ± 1°C). repetitive 12h light/dark cycles) for 1-week for acclimatization before administration of the CCl4. Normal diet of Sattu was given during this time as well. After CCl4 injection the mice were again kept for 1-week before drug administration. After which again a 1-week period was given before dissection.



Fig 9: Mice kept in cages for acclimatization

- The mice were randomly divided into equal experimental groups and were injected with CCl4(5 units) to induce hepatotoxicity.
- The drugs (both chemical and plant extract) were orally administered to the mice.

4.5.1 Dosage of Drugs:

- **Metadoxin**: 250mg, 500mg, 750mg to different experimental groups.
- **NAC**: 300mg, 600mg, 750mg to different experimental groups.
- **Neem**: 0.1g, 0.2g, 0.4g to different experimental groups.
- **Burdock Root:** 0.1g/ml, 0.2g/ml, 0.4g/ml to different experimental groups.

4.5.2 Preparation of Neem Extract:

Neem extract was prepared using neem leaves procured from local neem trees. The leaves were dried in hot air oven for 1-2 hours. Then they were kept in the open to dry for 24 hours until the leaves were crispy dry. The dry weight was noted. The leaves were then dissolved in ethanol. It was kept until the leaves were completely dissolved in ethanol. The solution was then heated until the ethanol evaporated completely such that the neem extract could be scraped from the utensil. The extract was then weighed according to the dosage and dissolved in edible vegetable oil before administering to the mice.

4.5.3 Dilution of Burdock root extract.

Burdock root extract was bought from brand Tonga Herbs. It was dissolved in water according to the dosage and then administered to the mice.

4.6 Parameters studied:

4.6.1Total Activity of Acid and Alkaline Phosphatases:

4.6.1.1 PNP Standard Curve:

The stock solution of Para-nitrophenol (PNP) was taken at a concentration of 0.5 mg/L and was diluted as 1/10, 2/10, 4/10, 6/10 and 8/10 concentration with double distilled water. Absorbance was measured at 440 nm and a standard curve was obtained.

4.6.1.2 Estimation of the total activity of Acid and Alkaline Phosphatases

For the estimation of the total activity of acid and alkaline phosphatases, the method of Bergmeyer et al. (1974) was followed with minor modifications. Enzyme activity was assessed from liver homogenate after the dissection of mice and kept on ice.



Fig 10: Liver homogeniser



Fig 11: Liver homogenate kept in ice

Acetate buffer (0.1M) at pH 5.0 and Glycine-NaOH buffer (0.1M) at pH 9.0 were used for the Acid phosphatase assay and the Alkaline phosphatase assay respectively. For the biochemical assay, the tissue homogenate was made to react with substrate (PNPP) and 0.1M Acetate/Glycine-NaOH buffer, followed by incubation at 37°C for 15 minutes. This was followed by the addition of 1 N NaOH. The absorbance was measured at 440 nm using an UV-spectrophotometer. Enzyme activity was determined from the PNP standard curve. The detailed experimental procedure is described in Annexure-II.

4.6.1.3 Assay Protocol:

System	Buffer (ml)	PNP (ml)	NaOH (ml)	Liver Homogenate(ml)	Incubation (min)	NaOH (ml)
Control	0.4	0.5	4	0.1	15	-
Experimental	0.4	0.5	-	0.1	15	4

4.6.2 Estimation of Alanine Aminotransaminase (ALT) activity from blood serum:

4.6.2.1 Mechanism of Action:

Alanine aminotransferase (ALT) catalyses the transamination of L-Alanine and α-Ketoglutarate (α-KG) to form Pyruvate and L-Glutamate. Pyruvate so formed is coupled with 2,4- Dinitrophenyl hydrazine (2,4-DNPH) to form a corresponding hydrazone, a brown coloured complex in alkaline medium and this can be measured colourimetrically. The colour intensity is proportional to the enzyme activity.

α -KG + L → Alanine Pyruvate + L-Glutamate

Pyruvate + 2,4DNPH → Hydrazone (Brown colour)

'Reitman and Frankel Method' is an end-point colorimetric method for the estimation of enzyme activity. To obtain accurate results, method has been standardized with kinetic method (Standard Karmen Unit assay). This Product is single point calibration version of original method to maximize ease of use and convenience. In this project work, the SGPT kit [MBK GPT (ALT) 2,4-DNPH Assay] manufactured by ARKRAY Healthcare Pvt. Ltd. has been used. In this kit, four reagents were provided. These are mentioned below:

Reagent No.	Reagent
1	Buffered Alanine-α-KG Substrate, pH 7.4
2	2,4-DNPH Colour Reagent
3	Sodium Hydroxide, 4N
4	Working Pyruvate Standard, 8 mM (150 IU/L)

4.6.2.2 Working reagent preparation:

• In the kit, reagents 1, 2 and 4 were ready to use.

• Solution 1 was prepared by mixing 1 mL of reagent 3 (4N NaOH) with 10 mL of purified (distilled/RO) water.

4.6.2.3 Specimen Handling

• Blood was obtained by cardiac puncture after sacrificing the mice

• The blood was allowed to settle and then centrifuged at 5000 rpm for 5 minutes to obtain serum

• The serum was stored in 2 degrees Celsius until the assay was performed.

4.6.2.4 Assay Protocol

Reagent/Substrate	Reagent Blank	Test	Substrate	Control
Volume in mL				
Buffered Alanine-alpha-KG substrate (pH 7.4)	0.25	0.25	0.25	0.25
Serum	-	-	0.05	-
Working Pyruvate standard 8 mM (IU/L)	-	0.05	-	-
Mix well and incubate at 37 degrees Celsius for 30 minutes				
2,4-DNPH (Colour reagent)	0.25	0.25	0.25	0.25
Purified water	0.05	-	-	-
Serum	-	-	-	0.05
Mix well and incubate at room temperature for 20 minutes				
NaOH (1:10 diluted with water)	2.5	2.5	2.5	2.5
Mix well and Absorbance is measured against purified water in a colorimeter at 505 nm within 15 minutes				

4.6.2.5 Calculation:

SGPT Activity =

$$\frac{(\text{Absorbance of Test} - \text{Absorbance of Control})}{(\text{Absorbance of Standard} - \text{Absorbance of Reagent Blank})}$$

Remarks: The unit of SGPT activity is IU/L.

4.6.3 Chromosomal aberration and visible side effects:

To check the Cytotoxic effect of Metadoxin and NAC, Neem and Burdock Root, it was observed whether any chromosomal aberrations occurred or not. Bone marrow samples collected from the femur bone of the mice were used for this purpose. The sample was first suspended in hypotonic KCl solution (0.1N).



Fig 12: Bone Marrow flushed using KCl

Then the cloudy cell suspension was centrifuged and after discarding the supernatant, the pellet was resuspended in chilled aceto-methanol solution (1:1). The sample was dropped on the 50% alcohol-soaked slide. Then the slides were flame dried and stained with Giemsa stain. After the slides were prepared, they were observed under a microscope at 10X, 40X and 100X magnification. The detailed experimental procedure is listed in Annexure-III.

5. RESULTS:

5.1 PNP Standard Curve

5.1.1 Table for absorbance of PNP at various concentrations

Amount of PNP (mg)	Absorbance at 440 nm
0.0	0
0.2	0.198
0.4	0.346
0.6	0.493
0.8	0.621
1.0	0.746

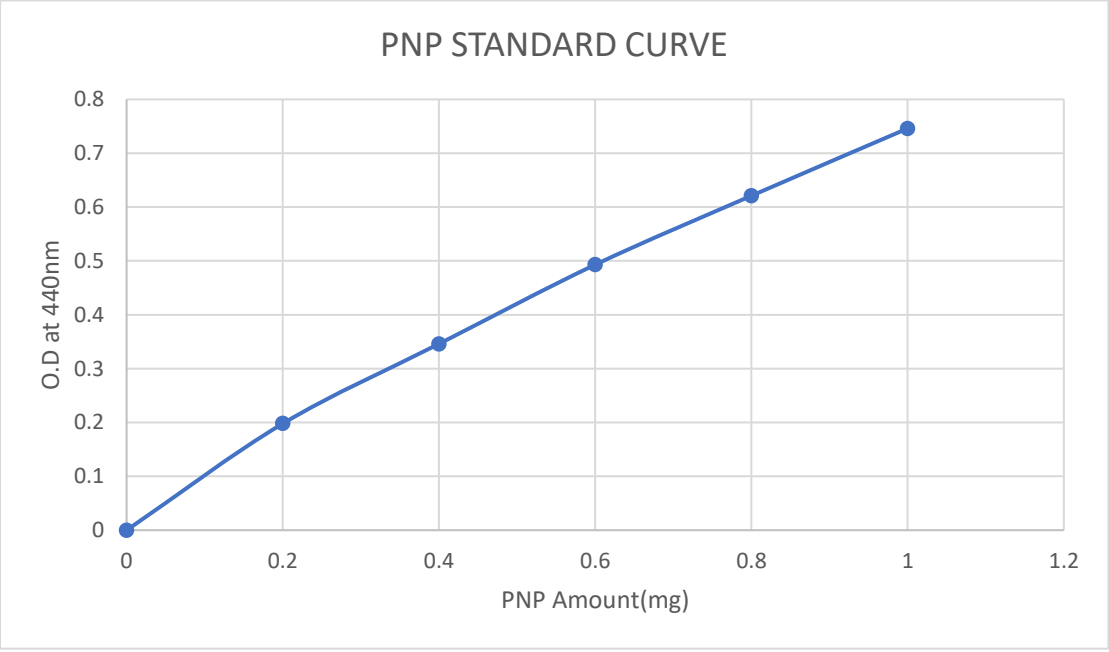


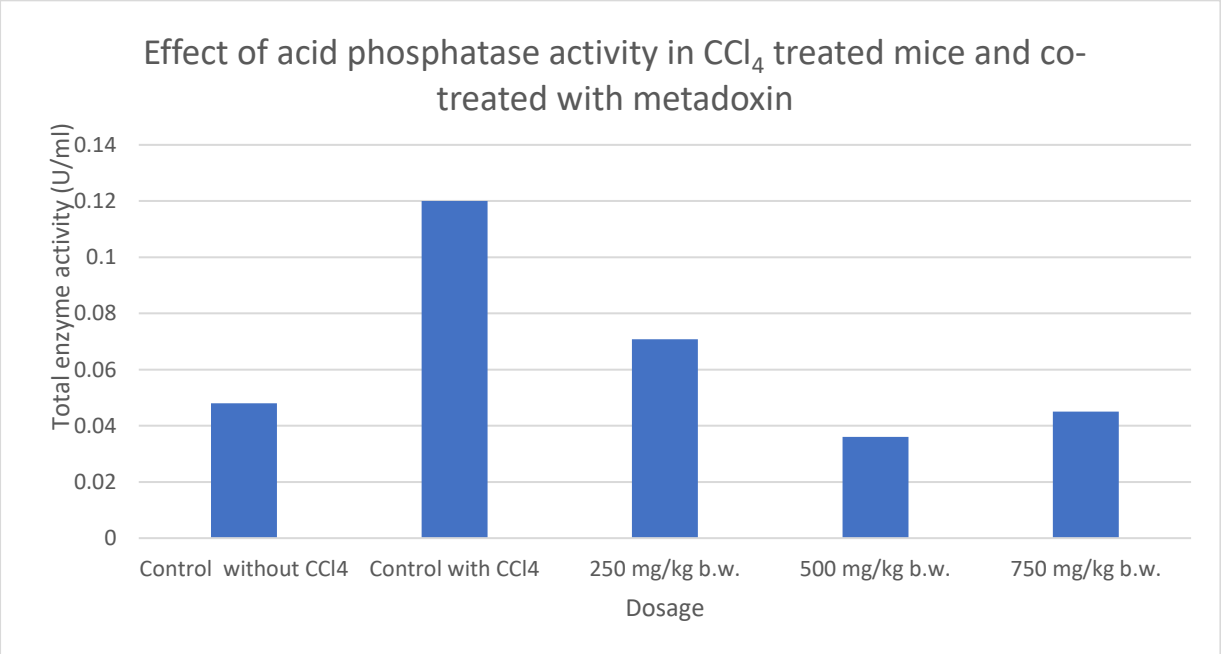
Fig. : PNP Standard Curve

5.2 Total Phosphatase Activity

5.2.1 Acid Phosphatase Activity Table for Metadoxine

Batch	Dose(mg/kg)	Enzyme Activity(U/mL)
Control	0	0.048
Control with CCl_4	0	0.05 ± 0.07
D1	250	0.0708 ± 0.0022
D2	500	0.036
D3	750	0.045 ± 0.007

Remarks: 1U = 1µg/min



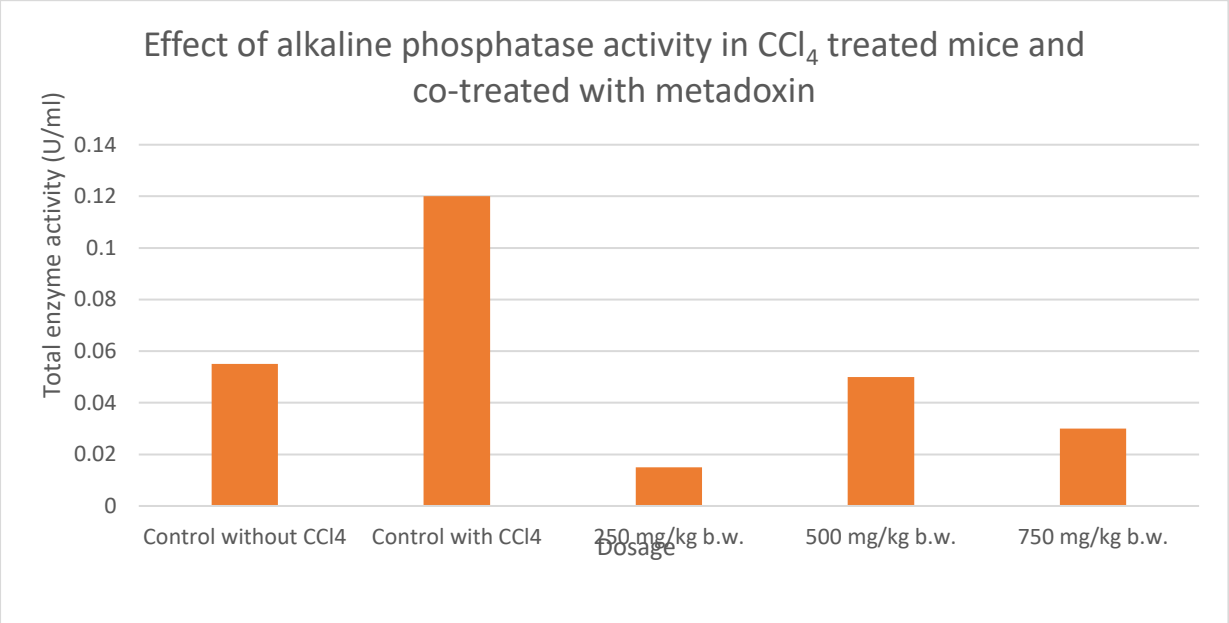
Major Findings:

When sub lethal doses of metadoxine was administered to the CCl_4 treated mice, it was seen that there was a substantial protective activity in the dose of 500mg/kg body weight, which was almost close to the control value without CCl_4. Enzyme activity in CCl_4 treated batch was close to 0.05U/ml.

5.2.2 Alkaline Phosphatase Activity Table for Metadoxine

Batch	Dose(mg/kg)	Enzyme Activity(U/mL)
Control	0	0.055
Control with CCl_4	0	0.031± 0.02
D1	250	0.015±0.001
D2	500	0.050
D3	750	0.03±0.014

Remarks: 1U = 1µg/min



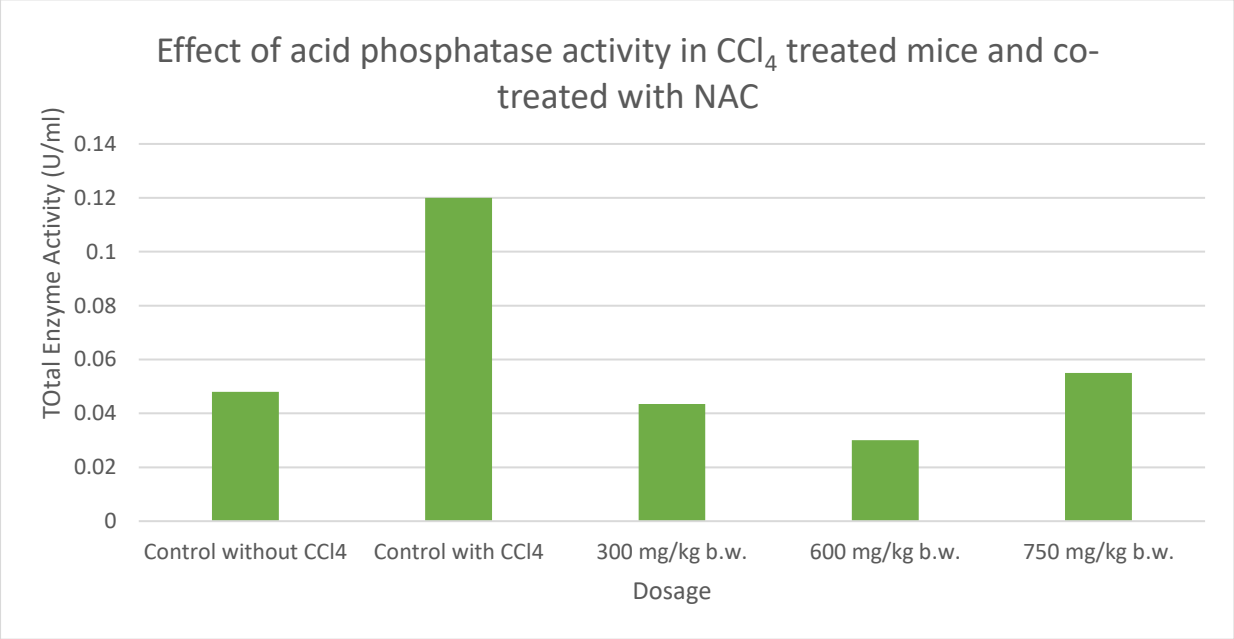
Major Findings:

When sub lethal doses of metadoxine was administered to the CCl_4 treated mice, it was seen that there was a substantial protective activity in the dose of 500mg/kg body weight, which was almost close to the control value without CCl_4. Enzyme activity in CCl_4 treated batch was close to 0.031U/ml.

5.2.3 Acid Phosphatase Activity Table for NAC

Batch	Dose(mg/kg)	Enzyme Activity(U/mL)
Control	0	0.048
Control with CCl_4	0	0.05± 0.07
D1	300	0.04347
D2	600	0.030
D3	750	0.055± 0.025

Remarks: 1U = 1µg/min



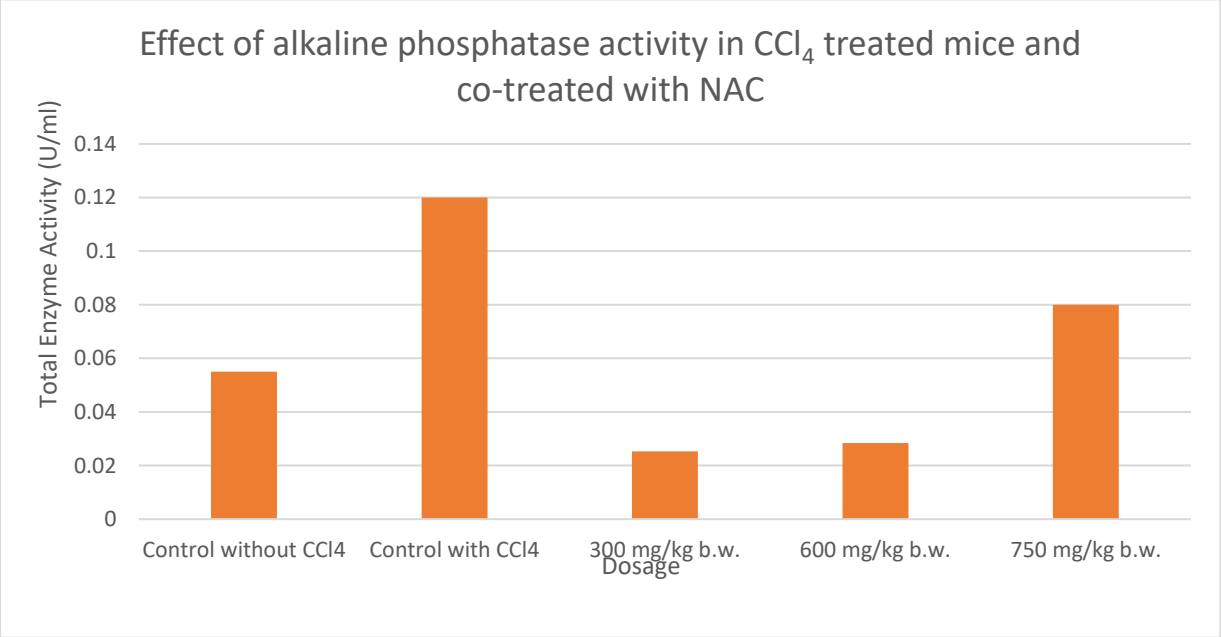
Major Findings:

When sub lethal doses of NAC was administered to the CCl_4 treated mice, it was seen that there was a substantial protective activity in the dose of 600mg/kg body weight, which was almost close to the control value without CCl_4. Enzyme activity in CCl_4 treated batch was close to 0.05U/ml

5.2.4 Alkaline Phosphatase Activity table for NAC

Batch	Dose(mg/kg)	Enzyme Activity(U/mL)
Control	0	0.055
Control with CCl_4	0	0.031 $\pm$0.02
D1	300	0.0253$\pm$0.027
D2	600	0.0284
D3	750	0.08

Remarks: 1U = 1µg/min



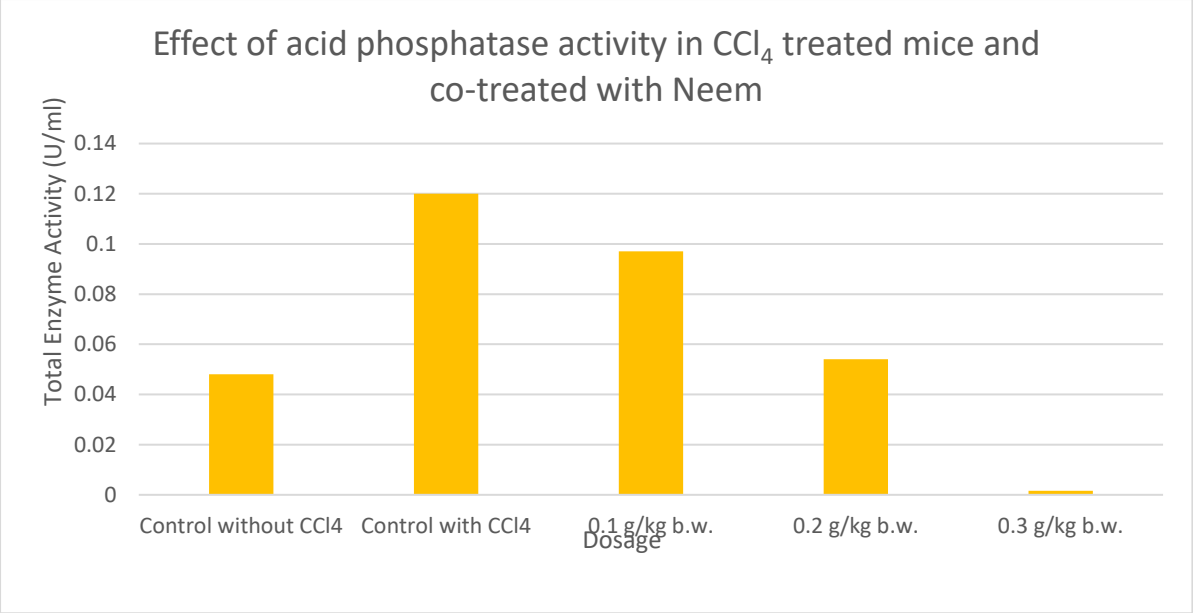
Major Findings:

When sub lethal doses of NAC was administered to the CCl_4 treated mice, it was seen that there was a substantial protective activity in the dose of 600mg/kg body weight, which was almost close to the control value without CCl_4. Enzyme activity in CCl_4 treated batch was close to 0.031U/ml.

5.2.5 Acid Phosphatase Activity Table for Neem

Batch	Dose(g/kg)	Enzyme Activity(U/mL)
Control	0	0.048
Control with CCl_4	0	0.05±0.07
D1	0.1	0.097±0.0273
D2	0.2	0.054±0.001
D3	0.3	0.00154

Remarks: 1U = 1µg/min



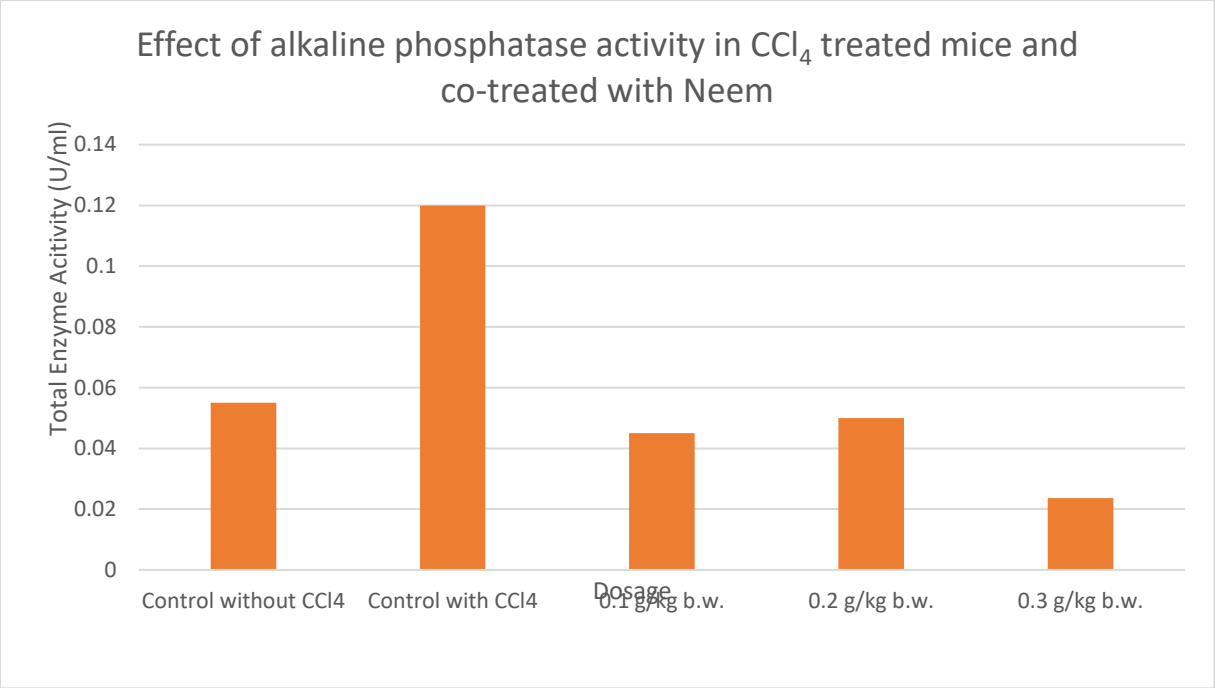
Major Findings:

When sub lethal doses of Neem extract was administered to the CCl_4 treated mice, it was seen that there was a substantial protective activity in the dose of 0.2g/kg body weight, which was almost close to the control value without CCl_4. Enzyme activity in CCl_4 treated batch was close to 0.05U/ml.

5.2.6 Alkaline Phosphatase Activity table for Neem

Batch	Dose(g/kg)	Enzyme Activity(U/mL)
Control	0	0.055
Control with CCl_4	0	0.031 ± 0.02
D1	0.1	0.045 ± 0.0041
D2	0.2	0.05 ± 0.00005
D3	0.3	0.0236

Remarks: 1U = 1µg/min



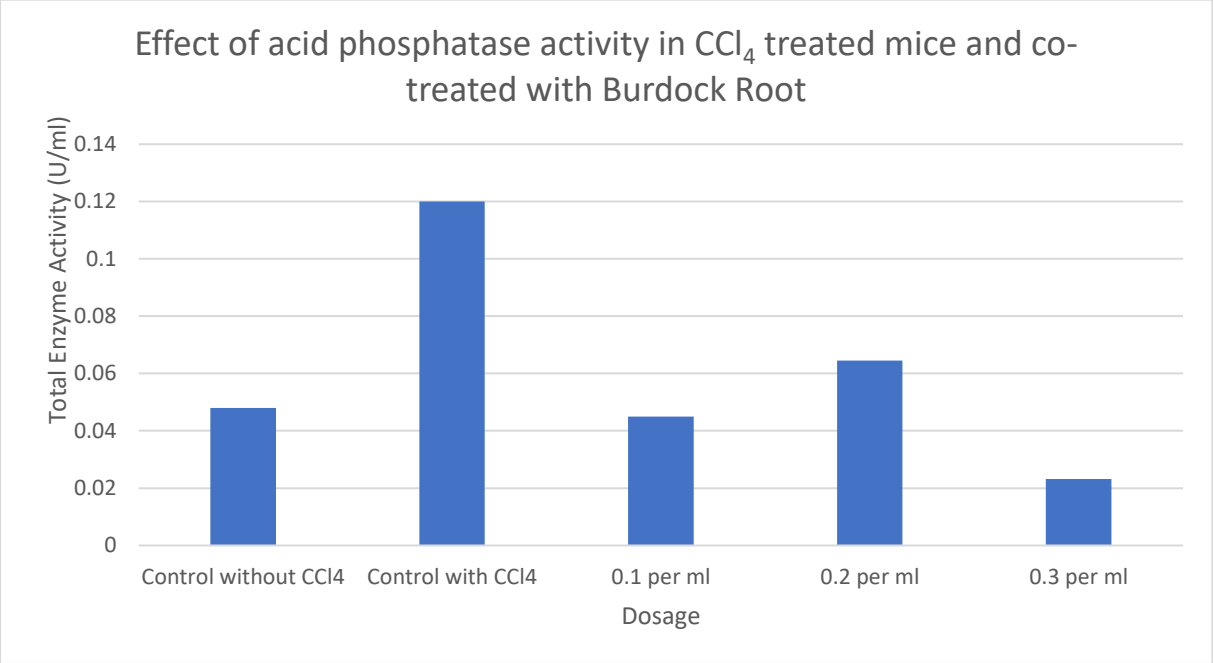
Major Findings:

When sub lethal doses of Neem extract was administered to the CCl_4 treated mice, it was seen that there was a substantial protective activity in the dose of 0.2g/kg body weight, which was almost close to the control value without CCl_4. Enzyme activity in CCl_4 treated batch was close to 0.031U/ml

5.2.7 Acid Phosphatase Activity Table for Burdock Root

Batch	Dose(per ml Water)	Enzyme Activity(U/mL)
Control	0	0.048
Control with CCl_4	0	0.05±0.07
D1	0.1	0.045±0.028
D2	0.2	0.0645±0.012
D3	0.3	0.0232±0.017

Remarks: 1U = 1µg/min



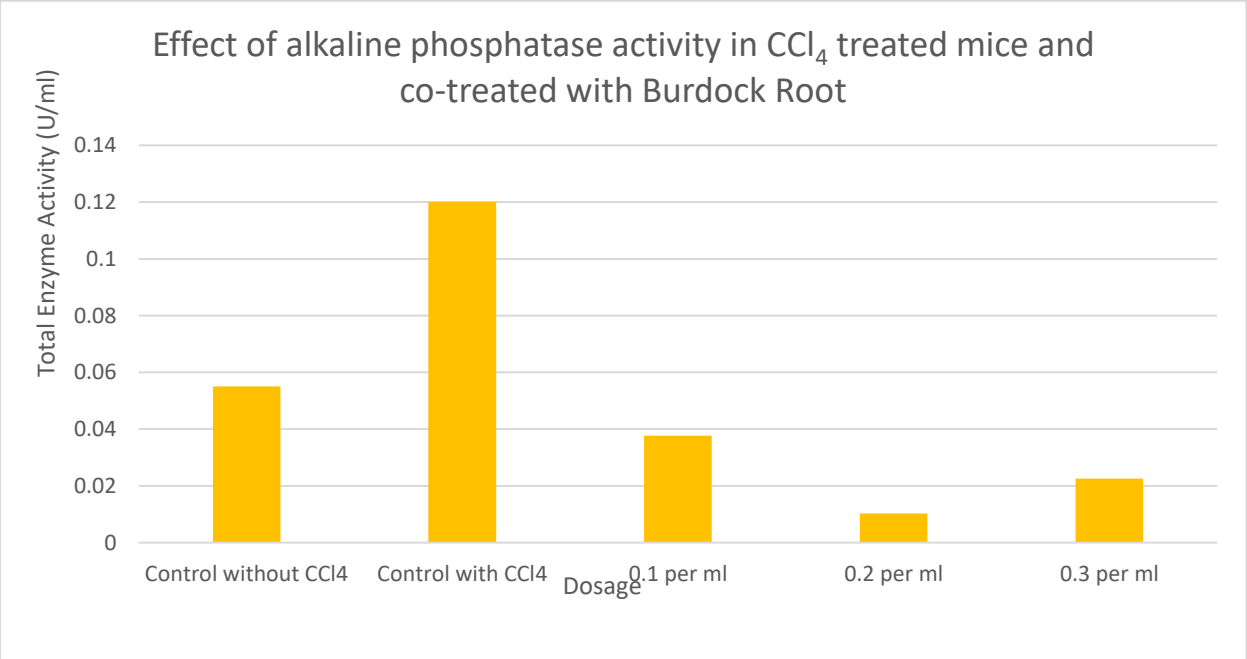
Major Findings:

When sub lethal doses of Burdock Root was administered to the CCl_4 treated mice, it was seen that there was a substantial protective activity in the dose of 0.1 per ml, which was almost close to the control value without CCl_4. Enzyme activity in CCl_4 treated batch was close to 0.05U/ml.

5.2.8 Alkaline Phosphatase Activity table for Burdock Root

Batch	Dose(per ml Water)	Enzyme Activity(U/mL)
Control	0	0.055
Control with CCl_4	0	0.031±0.02
D1	0.1	0.0377±0.0174
D2	0.2	0.01025±0.011
D3	0.3	0.02255±0.026

Remarks: 1U = 1µg/min



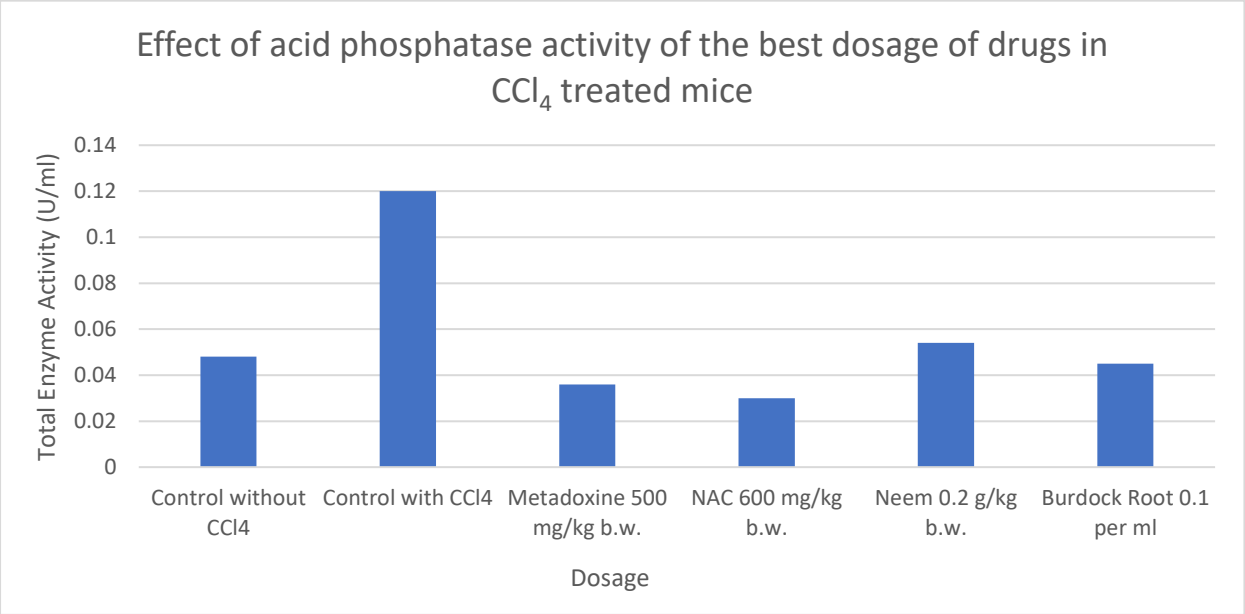
Major Findings:

When sub lethal doses of Burdock Root was administered to the CCl_4 treated mice, it was seen that there was a substantial protective activity in the dose of 0.1 per ml, which was almost close to the control value without CCl_4. Enzyme activity in CCl_4 treated batch was close to 0.031U/ml.

5.2.9 Acid Phosphatase Activity table for most effective dosage

Batch	Dose	Enzyme Activity(U/mL)
Control	0	0.055
Control with CCl_4	0	0.031±0.02
D1	Metadoxine 500 mg/kg b.w	0.036
D2	NAC 600 mg/kg b.w.	0.03
D3	Neem 0.2 g/kg b.w.	0.054
D4	Burdock Root 0.1 per ml	0.045

Remarks: 1U = 1µg/min



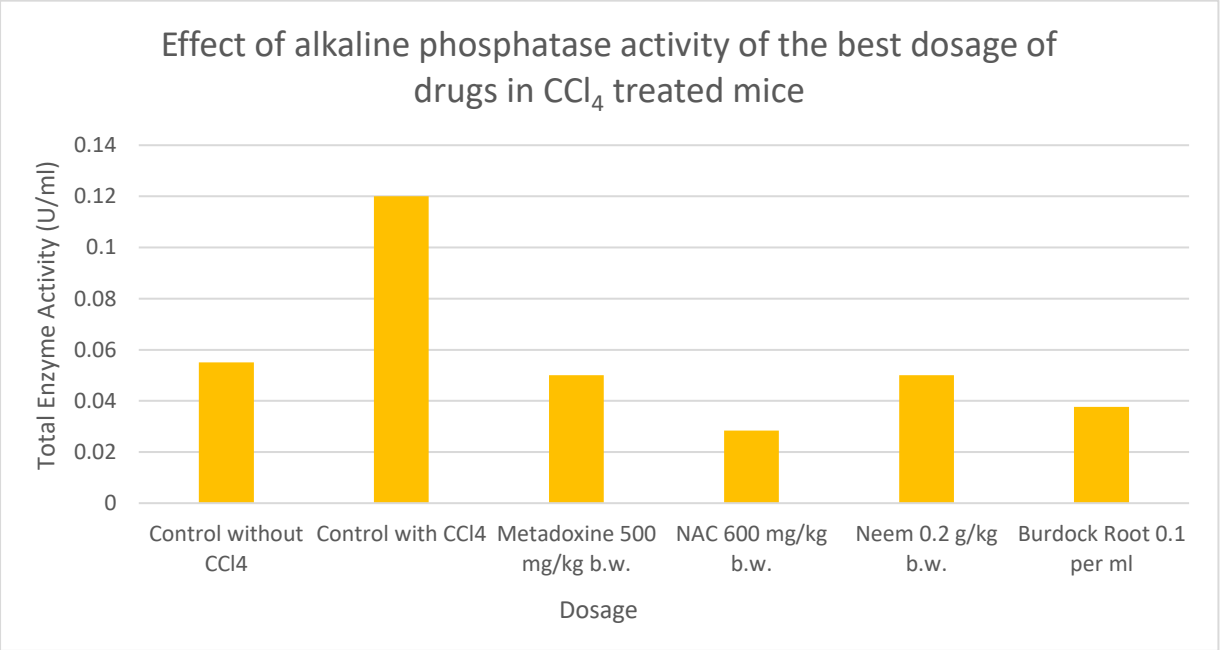
Major Findings:

When sub lethal doses of different drugs were administered to the CCl_4 treated mice, it was seen that there was a substantial protective activity in the dose of Neem(0.2 g/kg) body weight, which was almost close to the control value without CCl_4. Enzyme activity in CCl_4 treated batch was close to 0.031U/ml.

5.2.10 Alkaline Phosphatase Activity table for most effective dosage

Batch	Dose	Enzyme Activity(U/mL)
Control	0	0.055
Control with CCl_4	0	0.031±0.02
D1	Metadoxine 500 mg/kg b.w	0.05
D2	NAC 600 mg/kg b.w.	0.0284
D3	Neem 0.2 g/kg b.w.	0.05
D4	Burdock Root 0.1 per ml	0.0377

Remarks: 1U = 1µg/min



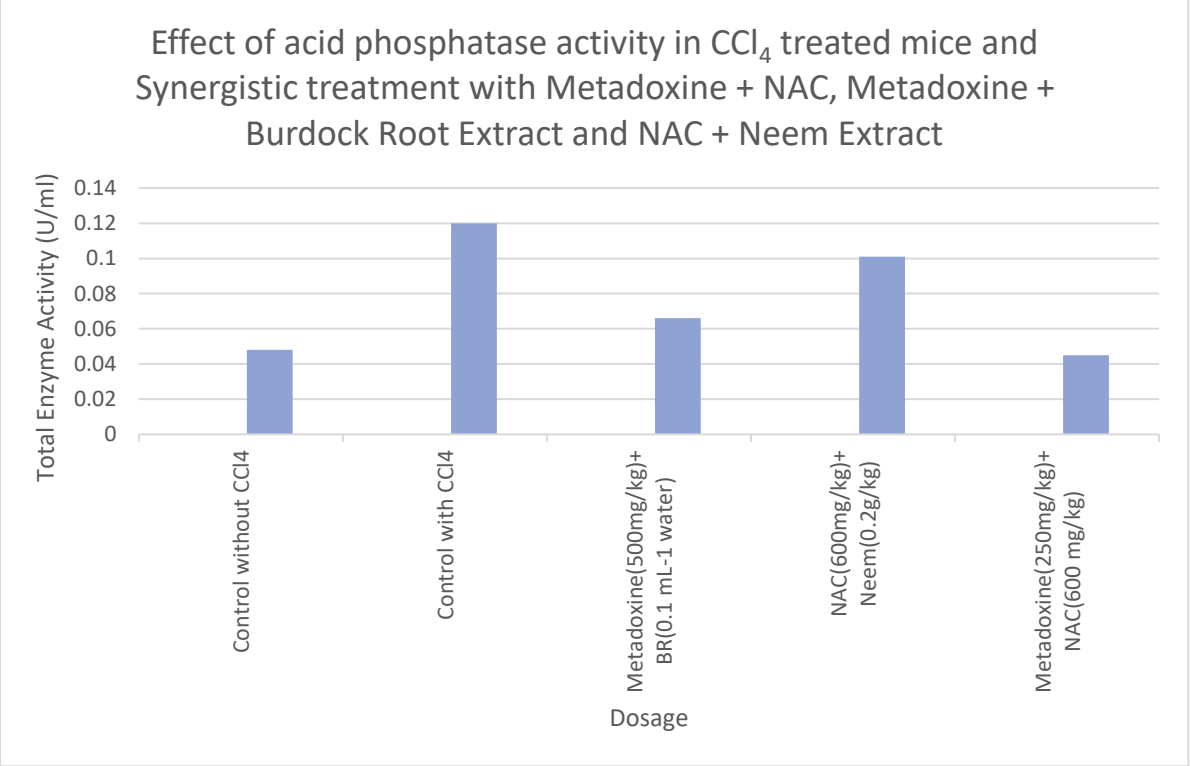
Major Findings:

When sub lethal doses of different drugs were administered to the CCl_4 treated mice, it was seen that there were substantial protective activity in the dioses of Neem(0.2 g/kg) body weight and Metadoxine(500mg/kg) body weight, which were almost close to the control value without CCl_4. Enzyme activity in CCl_4 treated batch was close to 0.031U/ml.

5.2.11 Acid Phosphatase Activity Table for Synergistic Drugs

Batch	Dose	Enzyme Activity(U/mL)
Control	0	0.048
Control with CCl_4	0	0.05±0.07
D1	Metadoxine(500mg/kg)+Burdock Root(0.1 mL^{-1} water)	0.066±0.0526
D2	NAC(600mg/kg)+ Neem(0.2g/kg)	0.101±0.106
D3	Metadoxine(250mg/kg)+ NAC(600 mg/kg)	0.045

Remarks: 1U = 1µg/min



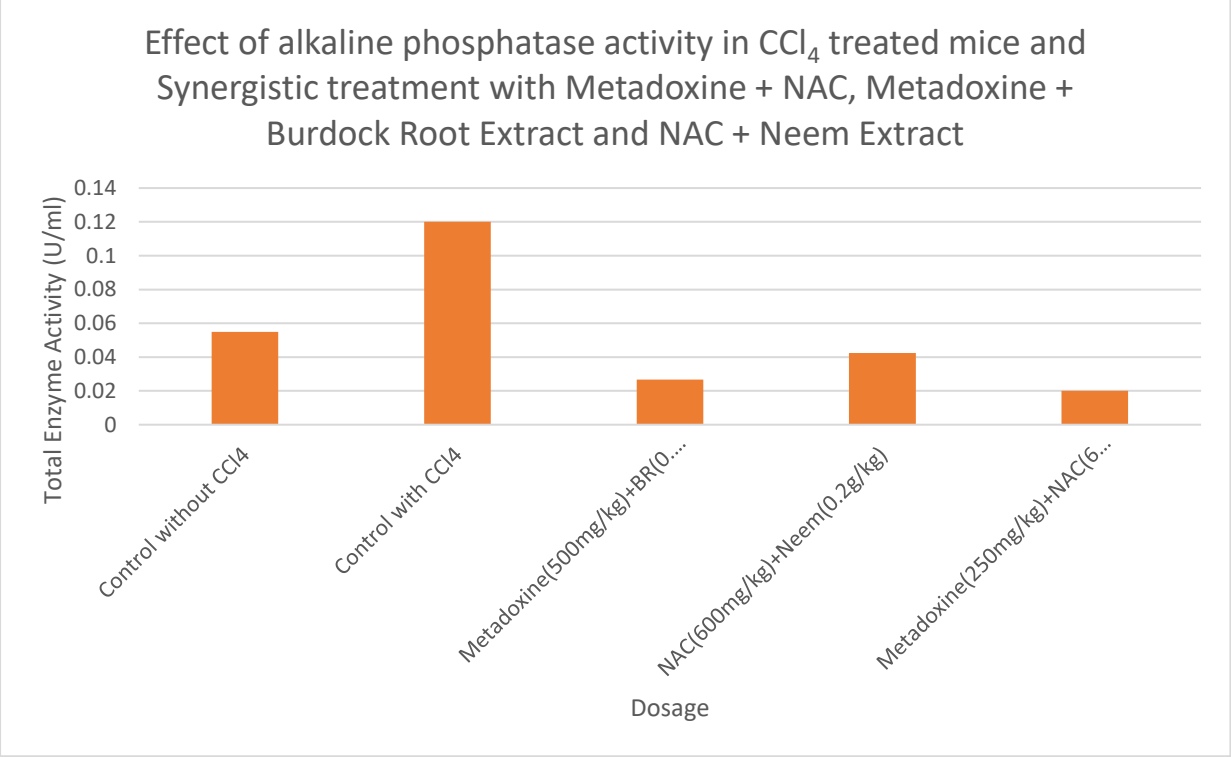
Major Findings:

When sub lethal doses of different Synergistic drugs were administered to the CCl_4 treated mice, it was seen that there was substantial protective activity in the dose of NAC(600 mg/kg) body weight + Metadoxine(250mg/kg) body weight, which was almost close to the control value without CCl_4. Enzyme activity in CCl_4 treated batch was close to 0.05U/ml.

5.2.12 Alkaline Phosphatase Activity Table for Synergistic Drugs

Batch	Dose	Enzyme Activity(U/mL)
Control	0	0.055
Control with CCl_4	0	0.05±0.07
D1	Metadoxine(500mg/kg)+Burdock Root(0.1 mL^{-1} water)	0.0267±0.0371
D2	NAC(600mg/kg)+ Neem(0.2g/kg)	0.0425±0.043
D3	Metadoxine(250mg/kg)+ NAC(600 mg/kg)	0.02

Remarks: 1U = 1µg/min

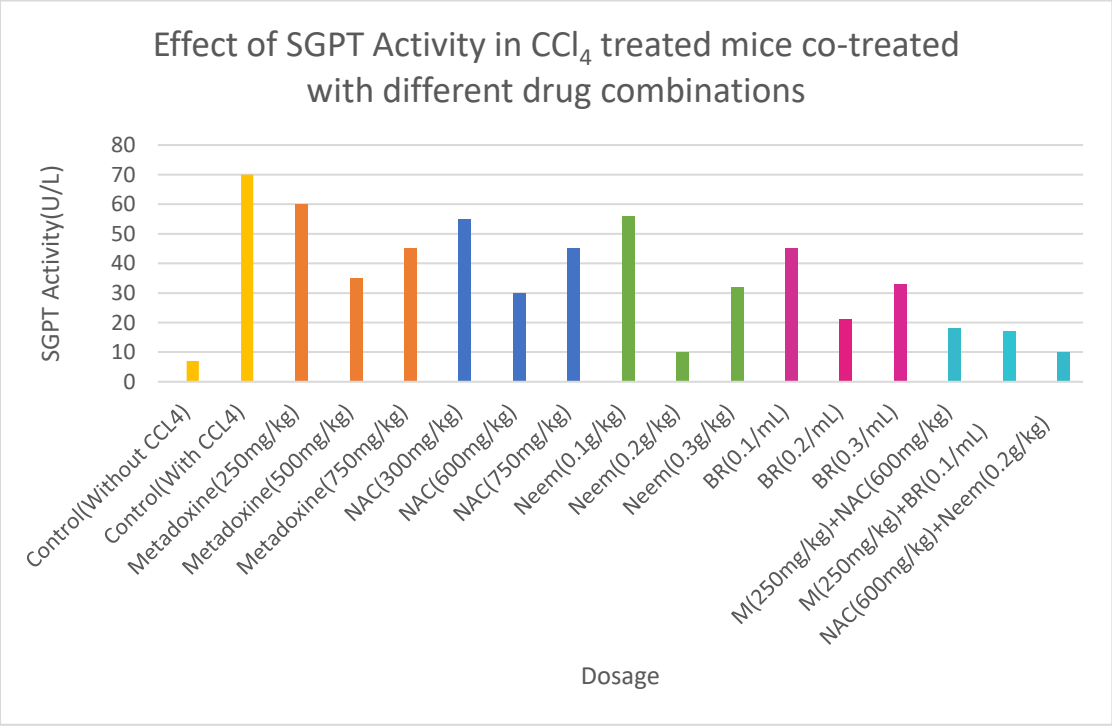


Major Findings:

When sub lethal doses of different Synergistic drugs were administered to the CCl_4 treated mice, it was seen that there was substantial protective activity in the dose of NAC(600 mg/kg) body weight + Neem extract(0.2g/kg) body weight, which was almost close to the control value without CCl_4. Enzyme activity in CCl_4 treated batch was close to 0.05U/ml.

5.3 SGPT Activity:

DOSE	SGPT Activity(U/L)
Control (Without CCL_4)	7
Control (With CCL_4)	70
Metadoxine(250mg/kg)	60
Metadoxine(500mg/kg)	35
Metadoxine(750mg/kg)	45
NAC(300mg/kg)	55
NAC(600mg/kg)	30
NAC(750mg/kg)	45
Neem(0.1g/kg)	56
Neem(0.2g/kg)	10
Neem(0.3g/kg)	32
BR(0.1/mL)	45
BR(0.2/mL)	21
BR(0.3/mL)	33
M(250mg/kg)+NAC(600mg/kg)	18
M(250mg/kg)+BR(0.1/mL)	17
NAC(600mg/kg)+Neem(0.2g/kg)	10

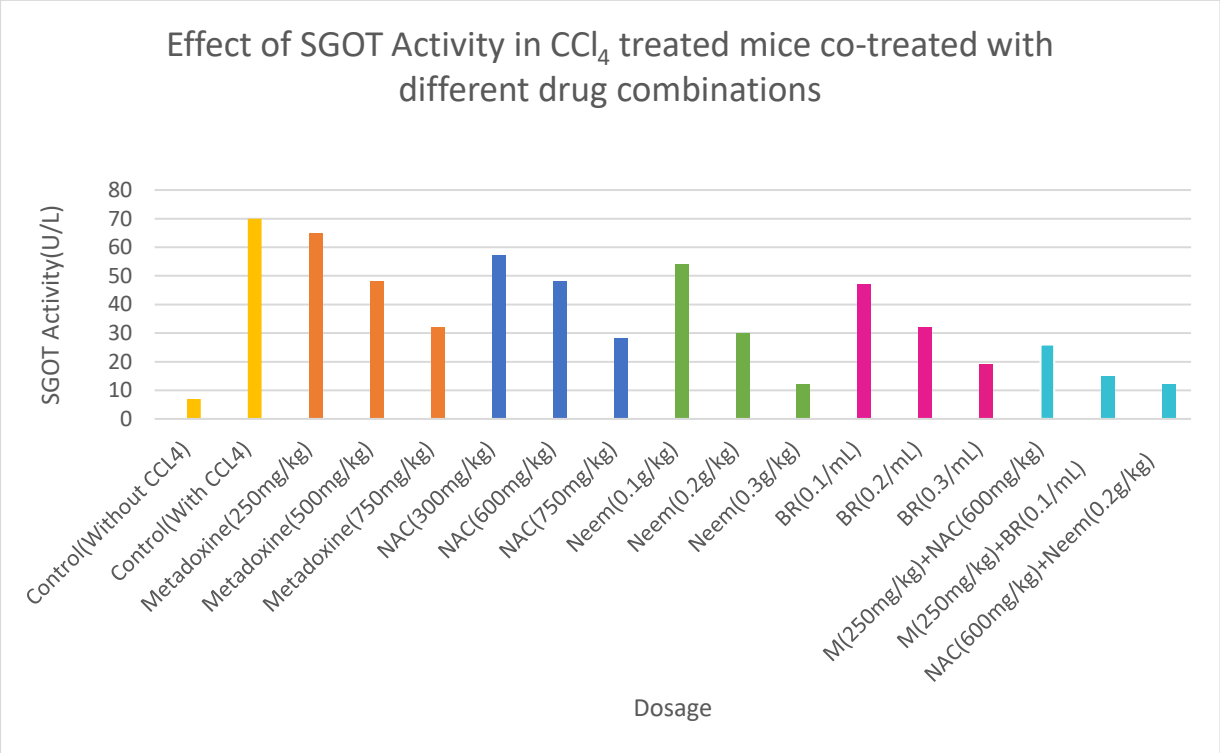


Major Findings:

When sub lethal doses of different drug combinations were administered to the CCl_4 treated mice, it was seen that there were substantial protective activity in the dioses of NAC(600 mg/kg) body weight + Neem extract(0.2g/kg) body weight and Neem extract(0.2g/kg) body weight, which were almost close to the control value without CCl_4. SGPT activity in CCl_4 treated batch was 70U/L.

5.4 SGOT Activity:

DOSE	SGOT Activity(U/L)
Control (Without CCL_4)	7
Control (With CCL_4)	70
Metadoxine(250mg/kg)	65
Metadoxine(500mg/kg)	48
Metadoxine(750mg/kg)	32
NAC(300mg/kg)	57
NAC(600mg/kg)	48
NAC(750mg/kg)	28
Neem(0.1g/kg)	54
Neem(0.2g/kg)	30
Neem(0.3g/kg)	12
BR(0.1/mL)	47
BR(0.2/mL)	32
BR(0.3/mL)	19
M(250mg/kg)+NAC(600mg/kg)	26
M(250mg/kg)+BR(0.1/mL)	15
NAC(600mg/kg)+Neem(0.2g/kg)	12



Major Findings:

When sub lethal doses of different drug combinations were administered to the CCl_4 treated mice, it was seen that there were substantial protective activity in the dioses of NAC(600 mg/kg) body weight + Neem extract(0.2g/kg) body weight and Neem extract(0.3g/kg) body weight, which were almost close to the control value without CCl_4. SGOT activity in CCl_4 treated batch was 70U/L.

5.5 Chromosomal Aberration Study

5.5.1 Chromosomal aberration study for Metadoxine:

Dose(mg/kg)	Normal Chromosome (%)	Pulverized Chromosome (%)
Control	92	7
Control with CCl_4	92	10
250	92	7
500	92	5
750	92	11

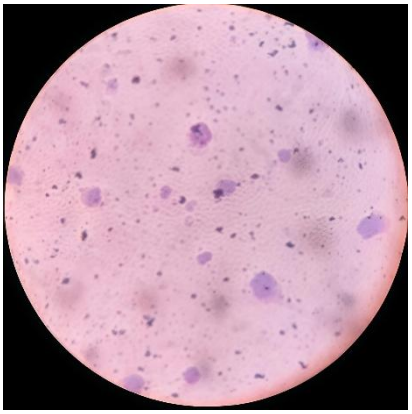


Fig 13: 40X magnified chromosome of Metadoxine administered mice

Major Findings:

When sub lethal doses of Metadoxine were administered to the CCl_4 treated mice, it was seen that there was a substantial decrease in the amount of pulverized chromosome in the dose of 250mg/kg body weight which generally indicates protective effect, and the value was almost close to the Control value without CCl_4. Pulverized chromosome amount in CCl_4 treated batch was 10% .

5.5.2 Chromosomal aberration study for NAC:

Dose(mg/kg)	Normal Chromosome(%)	Pulverized Chromosome (%)
Control	92	7
Control with CCl_4	92	10
300	92	8
600	92	9
750	92	10

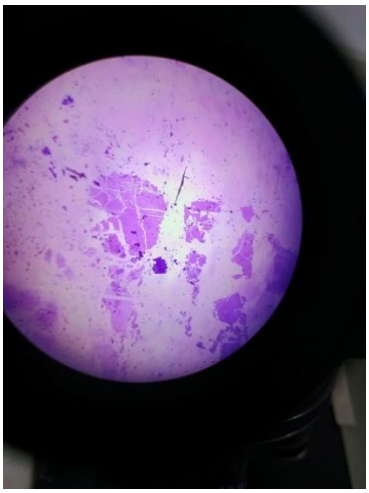


Fig 14: 40X magnified chromosome of NAC administered mice

Major Findings:

When sub lethal doses of NAC were administered to the CCl4 treated mice, it was seen that there was a substantial decrease in the amount of pulverized chromosome in the dose of 300mg/kg body weight which generally indicates protective effect, and the value was almost close to the Control value without CCl4. Pulverized chromosome amount in CCl_4 treated batch was 10% .

5.5.3 Chromosomal aberration study for Neem:

Dose(g/kg)	Normal Chromosome (%)	Pulverized Chromosome (%)
Control	92	7
Control with CCl_4	92	10
0.1	92	8
0.2	92	7
0.3	92	10

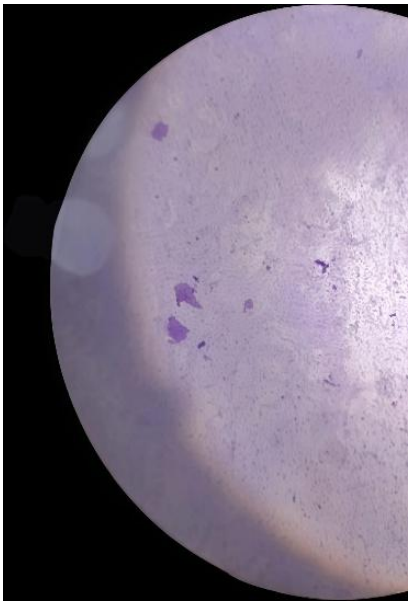


Fig 15: 40X magnified chromosome of Neem administered mice

Major Findings:

When sub lethal doses of Neem extract were administered to the CCl_4 treated mice, it was seen that there was a substantial decrease in the amount of pulverized chromosome in the dose of 0.2g/kg body weight which generally indicates protective effect, and the value was almost close to the Control value without CCl_4. Pulverized chromosome amount in CCl_4 treated batch was 10% .

5.5.4 Chromosomal aberration study for Burdock Root:

Dose(mL^{-1} water)	Normal Chromosome(%)	Pulverized Chromosome (%)
Control	92	7
Control with CCl_4	92	10
0.1	92	7
0.2	92	11
0.3	92	6

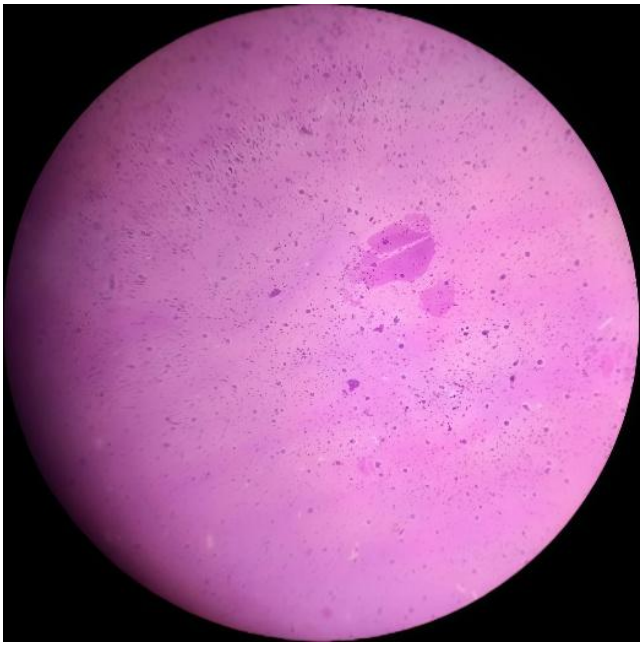


Fig 16: 40X magnified chromosome of Burdock root administered mice

Major Findings:

When sub lethal doses of Burdock Root were administered to the CCl_4 treated mice, it was seen that there was a substantial decrease in the amount of pulverized chromosome in the dose of 0.1 mL^{-1} water which generally indicates protective effect, and the value was almost close to the Control value without CCl_4. Pulverized chromosome amount in CCl_4 treated batch was 10% .

5.5.5 Chromosomal aberration study for Synergistic Drugs:

Dose	Normal Chromosome(%)	Pulverized Chromosome (%)
Control	92	7
Control with CCl_4	92	10
Metadoxine(250mg/kg)+ Burdock Root(0.1 mL^{-1} water)	92	11
NAC(600mg/kg)+ Neem(0.2g/kg)	92	11
Metadoxine(250mg/kg)+ NAC(600 mg/kg)	92	7



Fig 17: 40X magnified chromosome of Synergistic drug administered mice

Major Findings:

When sub lethal doses of Synergistic Drugs were administered to the CCl_4 treated mice, it was seen that there was a substantial decrease in the amount of pulverized chromosome in the dose of Metadoxine(250mg/kg)body weight + NAC(600mg/kg)body weight which generally indicates protective effect, and the value was almost close to the Control value without CCl_4. Pulverized chromosome amount in CCl_4 treated batch was 10% .

6. DISCUSSION

Two of the major biochemical assays for quantifying liver hepatotoxicity are Acid & Alkaline Phosphatase along with SGPT & SGOT. Acid phosphatase and alkaline phosphatase enzymes break down Para nitrophenol phosphate into yellow coloured Para nitrophenol and shows an elevated activity in CCl_4 induced hepatotoxic mice. Liver enzymes are increased many folds when liver is damaged, and this effectively gives measure of the amount of liver hepatotoxicity during the period of drug administration. Thus, the optimized doses, of the hepatoprotective agents which, lowers down Acid and Alkaline phosphatase activity close to the control activity can be considered to be most effective doses.

SGPT or ALT is an enzyme found in liver which catalyses the transfer of an amino group from alanine to α-ketoglutarate, forming pyruvate and glutamate. SGOT or AST is an enzyme found in liver which catalyses the transfer of an amino group from aspartate to α-ketoglutarate, forming oxaloacetate and glutamate. Elevated levels of these enzymes are an effective measure to estimate liver hepatotoxicity. The use of drugs Metadoxine, NAC and herbal extracts Neem and Burdock root have been seen to significantly reduce the levels of this enzyme.

Metadoxine accelerates ethanol metabolism thereby enhancing liver function. NAC on the other hand restores glutathione level to reduce oxidative damage and restore the enzyme levels. Neem acts as an antioxidant to reduce oxidative stress of liver cells. Burdock roots extract helps in Neutralizing reactive oxygen species and thereby acting as an anti-inflammatory agent.

Studies in the past two decades have revealed that administering a herbal extract along with a traditional chemical drug has improved the curing rates in patients with severe diseases. Using Piperine along with Rifampicin in Tuberculosis patients showed increased bioavailability of the drug. [Badmaev et al. 1999] In cancer studies also this is seen with many herbal extracts. Curcumin helps in enhanced anticancer activity when administered with Paclitaxel. [Goel et al. 2008] Reduced hepatotoxicity and improved efficacy is seen when Silymarin is administered along with Chemotherapy agents. [Kren & Walterova 2005] And also a very recent Nobel winning discovery was that when Artemisinin is used with antimalarial drugs to treat malaria it showed a very effective parasite clearance. [Tu Y. 2011]

Likewise, in this project we also aimed in finding out whether using any herbal extract enhances the effect of the chemical drugs Metadoxine and NAC or not. Hence, we chose the herbal extracts of Neem and Burdock both of which are known to have anticancer effects and in particular to combat liver hepatotoxicity. Subsequent studies have showed desired results.

7. CONCLUSION

- The collected data consistently demonstrate that **CCl_4 induces hepatic and cytogenetic toxicity**, as evidenced by:
  - **Reduced enzyme activity levels**, indicating impaired liver function.
  - **Increased pulverized chromosomes**, reflecting chromosomal damage.

Among the various **doses and treatment strategies** tested:

i. **Moderate doses** (e.g., **0.2 g/kg** or **500–600 mg/kg**) often showed **optimal protective effects**, restoring enzyme activity and reducing chromosomal aberration.
ii. **Very high doses** sometimes failed to enhance protection and, in some cases, **worsened toxicity**, suggesting a **non-linear dose-response**.
iii. Combination therapies revealed that:
   - **Metadoxine + NAC** was the **most consistently effective**, fully restoring both enzyme levels and chromosomal stability.
   - **Metadoxine + Burdock Root** also effectively restored enzyme levels close to normal and **NAC + Neem** was comparatively **less effective**.
   - The synergistic studies can also be conducted further using different dosage combinations to find the most effective combination.

Overall, **CCl_4 toxicity was reliably mitigated** by certain treatments, especially **Metadoxine + NAC**, and optimal individual doses. However, **not all combinations or high doses are beneficial**, highlighting the importance of **dose optimization and synergistic pairing** in therapeutic strategies against liver and genetic damage.

8. FUTURE SCOPE

- Dose Optimization and Mechanistic Studies:
  - Future work should explore the **optimal dose ranges** for each compound and combination, to prevent toxicity and maximize protective effects.
  - Investigating the **underlying mechanisms** (e.g., antioxidant pathways, DNA repair modulation, enzyme regulation) will help clarify **how** these agents confer protection.

- Long-Term Safety and Efficacy:
  - Conduct **chronic exposure studies** to assess the **long-term safety** and effectiveness of promising treatments like **Metadoxine + NAC**.
  - Evaluate **potential side effects** of higher doses or prolonged use.

- Broader Biomarker Evaluation:
  - Include additional **biochemical and genetic markers** (e.g., liver enzymes like ALT/AST, oxidative stress markers, comet assay) to **strengthen the findings**.
  - Use histopathological analysis for **tissue-level validation**.

- Comparative Studies with Standard Drugs:
  - Compare the efficacy of these combinations with **standard hepatoprotective drugs** (like silymarin or N-acetylcysteine alone) to position them in a clinical or pharmacological context.

- Translation to In Vivo and Clinical Models:
  - Extend studies to **larger animal models** and eventually consider **clinical trials** if safety and efficacy are confirmed.
  - Test applicability in other models of **toxic liver injury or genotoxicity** beyond CCl_4.

- Formulation and Delivery Research:
  - Explore **novel delivery systems** (e.g., nanoparticles, liposomes) to enhance the bioavailability of herbal or combined treatments.

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Annexure-I

PNP STANDARD CURVE

Introduction:

4 Nitrophenol (also called p-nitrophenol or 4-hydroxynitrobenzene) is a phenolic compound that has a nitro group at the opposite position of the hydroxyl group on the benzene ring.

Materials required:

• PNP (Merck, USA)

• 1N NaOH

• Double distilled water

• Weighing balance (Sartorius, Germany)

• Pipettes (Tarsons, India)

• Visible spectrophotometer (Labman, USA)

• 15 ml centrifuge tube (Tarsons, India)

• Cuvette

Procedure:

1. The stock solution of Para-nitrophenol was taken at a concentration of 5 mg/l, and was diluted as 2/10, 4/10, 6/10, and 8/10 concentration with double distilled water.

2. Absorbance was measured at 440 nm and a standard curve was obtained

Protocol of PNP Standard Curve:

Sl.No	PNP(Ml)	H_20(mL)	NaOH(Ml)	O.D(440nm)
1	0	1.0	4	0
2	0.2	0.8	4	0.198
3	0.4	0.6	4	0.346
4	0.6	0.4	4	0.493
5	0.8	0.2	4	0.621
6	1.0	0	4	0.746

Annexure-II

PHOSPHATASE ASSAY

Introduction: Phosphatases are enzymes that hydrolyze phosphate groups from a wide variety of organic substrates (phosphate esters), producing alcohol and phosphoric acid. They are found in all cells and usually are classified as either acid phosphatases or alkaline phosphatases. Various biological phenomena, such as proliferation, development, differentiation, and apoptosis are regulated by protein phosphorylation. Phosphorylation regulates the function, localization, and binding specificity of target proteins. Intracellular phosphorylation signaling is constructed based on the balance between phosphorylation by protein kinases and dephosphorylation by protein phosphatases (Sugiyama and Yamashita, 2016). Hence, it is imperative to quantify the enzymatic activity of hepatic phosphatase (both acidic and alkaline phosphatase) to look for probable toxicity owing to the administration of $CCl_4$to the mice at different doses.

In this experiment, p-nitrophenyl phosphate will be used to study the activity of the phosphatase enzyme. One of the products of the reaction, p-nitrophenol, turns yellow under basic conditions and can be detected using a spectrophotometer. The reaction is as follows:

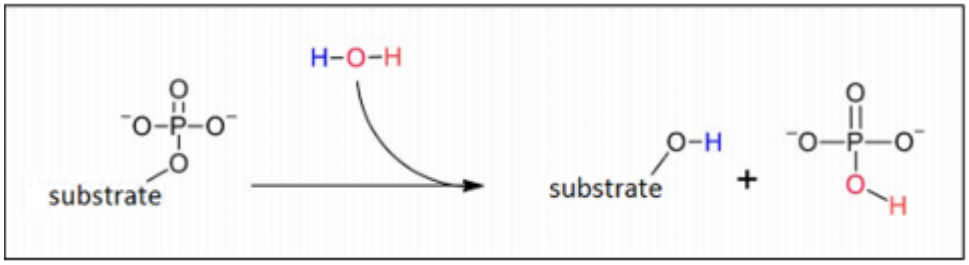


Fig. 18: Reaction mechanism of Phosphatase enzyme

Materials required:

I. Animal Model: Swiss Albino Mice (Mus musculus)

II. Chemicals and Equipment:

• Chloroform (Merck, USA)

• Metadoxine, NAC

• Acidic Buffer: 0.1 M Acetate buffer (pH=5)

• Alkaline Buffer: 0.1 M NaOH - Glycine buffer (pH = 9)

• Double distilled water

• 0.1 M 4-Nitrophenyl phosphate (PNPP) (Himedia, India)

• 4-Nitrophenol (PNP) (Merck, USA)

• I N NaOH

• Weighing balance (Sartorius, Germany)

• Pipettes (Tarsons, India)

• Petri dish

• Manual tissue homogenizer (Borosil, India)

• 15 ml Centrifuge tubes (Thermo Fisher Scientific, Pittsburgh)

• pH meter

• Centrifuging machine (REMI, India)

• Visible spectrophotometer (Labman, USA)

Procedure:

1. Mice were acclimatized in a laboratory environment for 7 days in dark - light cycle with regular feeding.

2. For each treatment dose, mice were euthanized using chloroform, and their weights were noted.

3. The liver was dissected, weighed and equally distributed respectively in acidic and alkaline buffers.

4. Extracted liver was homogenized in the respective buffer (0.24 g fresh tissue/ml buffer) solution using a manual tissue homogenizer

5. The semi-solid paste was poured into a centrifuge tube and kept on ice.

6. The tube was centrifuged at 5000 rpm for 5 min at room temperature.

7. The supernatant (Liver homogenate) was retained and the pellet was discarded.

8. Absorbance was recorded at 440 nm in a visible spectrophotometer.

9. A standard curve was prepared for 4-nitrophenol (PNP) using a standard protocol.

10. Total activity was determined from the standard curves of PNP and expressed as a microgram of PNP produced per mg per minute.

Assay Protocol:

System	Buffer (mL)	PNPP (mL)	NaOH (mL)	Liver Homogenate(mL)	Incubation	NaOH (mL)
Control	0.4	0.5	4	0.1	15	-
Experimental	0.4	0.5	-	0.1	15	4

Annexure-III

CHROMOSOMAL ABERRATION (CYTOTOXICITY)

Introduction: Chromosomes are heritable material inside the cell that has unique properties that distinguish them from other cellular organelles. The number of chromosomes varies between different species but remains constant from cell to cell within the same organism. Chromosomes play an important role during cell division, which ensures that DNA replication is accurate and distributed between newly formed cells after cell division (Jain et al., 2017). In essence, structural chromosome aberrations are the result of chromosome breakage and abnormal reunion of broken chromosomes.

Materials required:

I. Animal Model: Swiss Albino Mice (Mus musculus)

II. Chemicals and equipment:

• Metadoxine, NAC

• Chloroform (Merck, USA)

• Hypotonic solution: 0.01 M Potassium Chloride (KCl) solution

• Fixative: Chilled aceto-methanol (1:1)

• Giemsa stain (Merck, USA)

• 50% chilled ethanol (Merck, USA)

• Double distilled water

• Ice

• Weighing balance (Sartorius, Germany)

• Hypodermic syringe

• Forceps

• Scissors

• Petri dish

• Microscope (Optical; Magnus opto)

• 15 ml centrifuge tube (Tarsons, India)

• Pasteur pipettes

• Grease-free glass slides

• Coverslips

• Spirit lamp

• Refrigerator

• Centrifuge machine (Remi, India)

• Pasteur pipettes

Procedure:

1. Mice were acclimatized in a laboratory environment for 7 days in dark light cycle with regular feeding

2. Mice were injected (IP) with Metadoxine and NAC at different doses.

3. Bone marrow was collected from long bones (humerus and femur). Femur was dissected out sacrificed with Chloroform.

4. The tubular bone was flushed out with the hypodermic syringe. It was kept for 20 minutes (optimum time for swelling of cells) in an incubator at 370C (for better permeability).

5. KCl hypotonic solution and the marrow was collected in a centrifuge tube and gently flushed to make a cloudy cell suspension.

6. The material was centrifuged in a centrifuge machine at 5000 rpm for 5 minutes at room temperature.

7. The pellet deposition formed at the bottom of the tube was kept for further use and the supernatant was discarded.

8. The pellet was fixed in chilled aceto-methanol (1:3), flushed gently and kept for 20 minutes for proper fixation.

9. After fixation, the material was centrifuged again at the same speed for 5 minutes at room temperature.

10. The supernatant was discarded, and the pellet was collected and resuspended in fixative and kept overnight.

11. Slides were made grease-free and stored overnight in chilled 50% alcohol.

12. The pellet was centrifuged once again, the supernatant was discarded and fresh fixative was added to the pellet.

13. It was flushed gently with a Pasteur pipette.

14. 2 drops of cell suspension were dropped on the 50% alcohol-soaked slide.

15. The slide was burnt lightly on a spirit lamp flame and air-dried (so that the cells swell and stick on the slide).

16. Air-dried slides were stained in Giemsa stain for 45 minutes under covered condition.

17. The slides were then washed gently in running tap water and observed under a microscope at 10x followed by 40x magnification.

10. APPENDIX

DrillBit

The Report is Generated by DrillBit Plagiarism Detection Software

Submission Information

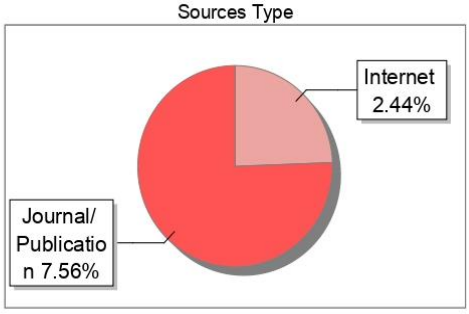
Author Name	Anuska sarkar
Title	In vivo studies of Metadoxine, NAC, Neem and Burdock Root on CCl4-induced Toxicity on Swiss Albino mice (Mus musculus)
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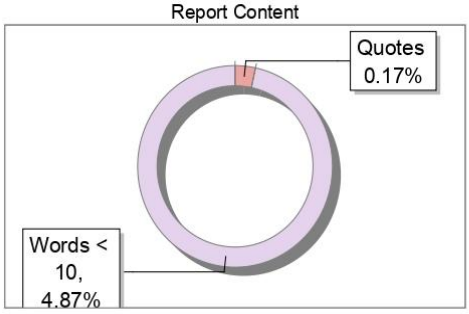
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